# Talking while Tran/s/: Transfeminine Negotiations of the Cisgender Listener

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A Thesis
Presented to
The Division of Philosophy, Religion, Psychology, and Linguistics
Reed College

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In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Arts

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Wren Arbor

May 2026

Approved for the Division
(Linguistics)

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# Acknowledgements

I want to thank a few people.

# List of Abbreviations

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| **CoG** | Center of Gravity; a measure in Hz of mean acoustic pitch at a given moment, weighted by amplitude. |
| **CP** | Conversation Partner; the term I use to describe the cisgender woman and transgender woman who played the role of the other side of each conversation the speaker participants had. |
| **HRT** | Hormone Replacement Therapy; a common medical intervention for gender transition. |
| **BIPoC** | Black, Indiginous, and People of Color. |

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# Abstract

Research on transgender speech has primarily been of two categories: aggregated comparisons against cisgender speakers, and ethnographic exploration within queer in-groups; neither of these address how trans speakers navigate cisgender perceptions. Frontness of the /s/ sound is a well-studied index of femininity and gay-masculinity within the gender norms of the cisgender white English-speaking population; I investigate how four transfeminine speakers' navigate these norms through a comparison of their /s/ pronuncitions with two listeners, one transgender woman and one cisgender woman, to fill this gap. I audio-recorded one-on-one casual conversations between speaker and each listener, acoustically comparing speakers' pronunciation of /s/ in a conversation with each listener. I then interviewed each speaker, assessing metalinguistic concerns and attitudes in three areas hypothesized to predict shifts: speakers concerned with cis perceptions were expected to defensively shift towards hegemonically normative feminine /s/ pronunciations, whereas speakers concerned with sounding gay-masculine or who distanced themselves from hegemonic femininity were expected not to. I find that two speakers shifted towards fronter /s/ pronunciations with the cis listener, and two speakers did not. Counter to my predictions, these speakers who shifted to fronter /s/ pronunciations were the two who expressed concerns about sounding like gay men, whereas all speakers were similarly concerned with cis perceptions, and all speakers except for one who did not shift to fronter /s/ pronunciations distanced themselves from hegemonic femininity, counter to my predictions. Reinterpreting my initial model, I consider /s/ fronting to hold greater salience for speakers concerned with sounding like gay men, both as an index of gay-masculinity and as an index of femininity, highlighting the interdependence of these indexical orders. These speakers take up the feminine meaning of /s/ fronting as a resource for self-defense against potentially-misgendering cisgender listening practices.

# Introduction

Along with the visual presentation of one's body, the sound of one's voice is among the most strongly gendered aspects of behavior. Transgender people[1] know this intimately, as our speech is highly scrutinized by those who – consciously or unconsciously – seek to enforce the gender binary (Jessee and Calder 2025). In this thesis, I investigate how this knowledge manifests by comparing English-speaking transfeminine[2] speakers' linguistic behavior across two conversational contexts: a conversation with a cisgender[3] woman, and a conversation with a trans woman. I focus on their pronunciation of the /s/[4], sound (the first sound in 'safe' and the last sound in 'face'), a well-studied site of gendered linguistic variation.

It has become widely accepted both in the academic study of gender and parts of the popular consciousness that gender is performative (Butler 1990; Henry and Hall 2025). Most simply, this means that gender does not exist outside of cultural and historical context, and is constantly constructed and reconstructed by social behavior. Under this view, scholars of gender develop their models by studying the behaviors involved in gender, and how those behaviors come to have gendering effects. Language is of primary importance to the construction of gender; speech about gender is arguably the foundation of gender's existence. For instance, the speech of medical specialists conducting an ultrasound or delivering a newborn – "it's a girl!" or "it's a boy!", or the imperatives "act ladylike" and "man up", do the performative work of enforcing and solidifying gender (Livia and Hall 1997b).

Yet, talk about gender does not represent the beginning and end of the linguistic performativity of gender. Sociolinguistics, the study of language and the social world, recognizes that variations in language use beyond the referential – less "what one says", more "how one says it" – are socially meaningful and per-

[1] I use 'transgender' or 'trans' to refer to those who identify with this label. Many transgender people understand the term broadly to include all who do not fully identify with the gender assigned to them at birth. However, the term is sometimes narrowed to refer mostly to transgender men and transgender women, who primarily identify with the opposite binary gender of the gender assigned to them at birth.

[2] I use the term 'transfeminine' to refer to thsoe who identify with this label, primarily trans women and feminine nonbinary people (especially those assigned male at birth, for whom transition is a move towards a feminine identity).

[3] I use 'cisgender' or 'cis' to refer to people who are not transgender.

[4] In this thesis, slashes denote that the enclosed text is written in the International Phonetic Alphabet. /s/ refers to the voiceless alveolar fricative speech sound. Although it looks the same as the English letter 's', this symbol refers to the sound, rather than the spelling (see International Phonetic Association 1999).

formative (Henry and Hall 2025). The common belief that one's gender can be guessed or determined from the sound of their voice attests to the importance of the sound of the voice in the construction and performance of gender. For example, if a perciever hears a voice that is higher pitched, they may assume the speaker is a woman, whereas if it is lower pitched, they may assume the speaker is a man. Here, the perciever construes high pitch to be a sign of femininity, and low pitch to be a sign of masculinity. Associations like these are known in sociolinguistics as 'indexical', as the relationship is imagined to be causal or co-occurent, much like the indexical link between smoke and fire.

It bears emphasis that the process of interpreting variations in language use (such as variations in voice pitch) as meaningful signs depends on perceivers' understanding of the world (Gal and Irvine 2019). Some indexicalities, such as the indexical link between voice pitch and gender, are the products of understandings of the world which align with and support structures of power (Calder 2021). These indexicalities come to be entrenched as expectations and norms, benefitting those who fit into gendered ideals and marginalizing those who do not. Although many cultures have somewhat overlapping gender ideals, including the idealization of the feminine voice as higher-pitched and the masculine voice as lower-pitched, the character of these indexicalities and their applications is nonetheless culturally specific. In the western, English-speaking, North American cultural context of this study, gendered indexicalities such as this one are embedded in an understanding of gender which idealizes those in power and privilege here: white, cisgender, heterosexual, middle-class men and women. These norms serve to marginalize those outside of this group, deeming their language use inadequate and inappropriate (eg. Inoue 2006; Inoue 2003; Flores and Rosa 2015; Heffernan 2023).

In investigating indexicality, existing sociolinguistic studies primarily focus on only one set of these expectations, either the normative, powerful set described above (eg. Hazenberg 2012; Podesva and Van Hofwegen 2015) and henceforth referred to as 'hegemonic', or alternative, local sets constructed within marginalized communities (eg. Mendoza 2024; Calder 2019a). Within the realm of the study of trans speech, the former type of study often interprets trans voices in their deviance from or compliance with cisnormativity, and runs the risk of tacitly enforcing hegemonic norms, marginalizing the diversity of trans voices (Calder and A. Steele 2025). Furthermore, focus within a single interpretive lens – hegemonic or otherwise – often sidelines the intersubjective nature of identity, by which identity emerges in the interaction of subjects and their establishment of contrast between self and other, speaker and listener (Bucholtz and Hall 2010). Marginalized speakers hold awareness both of hegemonic and alternative indexicalities, and must be aware of how their listener will interpret their speech.

Recent research has worked to take up this perspective, placing more even focus on both sides of the speaker-listener pair by investigating how trans voices are perceived by listeners (Jessee and Calder 2025) and trans people's understandings of these perceptions (Heffernan 2023). However, prior research has not investigated how these understandings manifest in trans speakers' actual linguistic practice. A mainstay of sociolinguistics is the study of changes within individual speakers'

language use in response to changes in context, known as style-shifting. Many of these studies investigate how speakers change their speech depending on their listener, finding listener or audience to be a primary motivating factor for shifts (e.g. Podesva 2007; Rickford and Mcnair-Knox 1994). A study of style-shifting on trans speakers treating the listener as a variable, has not yet been conducted. By filling this gap, I provide data on how transfeminine speakers navigate cisgender linguistic norms.

Much of the existing study on listener-oriented style-shifting primarily finds that speakers often change their linguistic behavior to be more similar to the speech of their listener, known as audience design, or accomodation (eg. Rickford and Mcnair-Knox 1994). I address accomodation in my discussion of my results, finding that it is certainly in play in my data. However, in line with contemporary models of style-shifting research (eg. Podesva 2007), I consider theories of accomodation to nonetheless provide an incomplete analysis of my data. In contrast, an interpretive model which understands speakers to be reacting to the trans status of their listener and to the structure of hegemonically-normative indexicalities provides a more full analysis.

Language and gender research has found that, similar to voice pitch, the English /s/ sound varies in pronunciation, and that hegemonic gender norms expect men and women to pronounce /s/ differently (Flipsen et al. 1999; Hazenberg 2012; Zimman 2013; Podesva and Van Hofwegen 2015). I focus on this sound for two primary reasons. First, unlike voice pitch, gendered variation in /s/ is not influenced by gendered anatomy; voice pitch is influenced by the size of the vocal tract, whereas /s/ pronunciation is not (Zimman 2017c). Second, research has found that when men pronounce /s/ differently than norms expect them to, their voices are percieved as sounding gay (e.g. Calder 2019a). For transfeminine speakers, this makes /s/ particularly difficult territory, as it is possible moving toward feminine /s/ pronunciations may only result in being mispercieved as gay men. In order to study how trans speakers navigate hegemonic linguistic norms, I look to this difficult situation, investigating how transfeminine speakers shift /s/ pronunciations from a conversation with a transgender listener to one with a cisgender listener, and what factors may influence the (conscious or unconscious) decision to shift.

There are three possible outcome behaviors: speakers shift towards more hegemonically feminine /s/ pronunciations with the cis listener, speakers do not shift, or speakers shift away from hegemonically feminine /s/ pronunciations with the cis listener. In the first case, I predict that speakers who are unconcerned with being mistaken for gay men but who do wish to avoid being judged as insufficiently feminine by cis listeners are most likely to shift towards more hegemonically feminine /s/ pronunciations with the cis listener. The other two outcomes (either no shift or a shift away from hegemonically feminine /s/ pronunciations with the cisgender listener) would be most likely for speakers who differ from this description: those who are concerned with being mistaken for gay men, those who are unconcerned with cisgender judgements.

In order to test these predictions, I recruited 4 transfeminine speakers and 2 women conversation partners, one trans and one cis. I conducted 20-30 minute

one-on-one audio recorded casual conversation sessions between each speaker and each conversation partner in which participants were asked to collaboratively play a word game and then to chat casually and get to know each other. I then acoustically measured their /s/ pronunciations in each recording. After the conversation sessions, I interviewed each speaker, asking questions about their relationship with gender, sexuality, and their voice, with a focus on the challenges of dealing with cisgender perceptions, and assessed the extent to which speakers expressed the concerns and goals pertaining to my predictions.

I find that two speakers shift towards more hegemonically feminine /s/ pronunciations with the cisgender listener, one speaker does not shift, and one speaker shifts away from hegemonically feminine /s/ pronunciations with the cisgender listener. Counter to my predictions, the former two speakers (who shifted towards more hegemonically feminine /s/ pronunciations) were the two who expressed concern about sounding like gay men, and the latter two speakers did not express these concerns. Also counter to my predictions, all speakers seemed similarly concerned with cisgender judgements, and the speaker who shifted away was the only one who did not particularly distance herself from hegemonic femininity.

In interpreting these results, I first consider whether my speakers' shifts are actually in reaction to whether their conversation partner is cisgender, rather than simply in accomodation to the way their conversation partner pronounces /s/. I am unable to conclude with certainty that the former is the case, but I nonetheless conclude that it is more likely for two reasons. First, my speakers emphasize the influence cisgender perceptions have on their relationships with their voices in my interviews with them, encouraging an interpretation which considers this influence. Second, the accomodation approach does not provide sufficient explanation as to why two speakers shifted /s/ in the predicted direction and two did not. In answering this question, I reinterpret my initial predictive factors, finding that these two groups are indeed differentiated by their expression of concerns with sounding gay-masculine. Rather than choosing not to shift to avoid misinterpretations of /s/ pronunciations as indexing gay-masculinity, I instead interpret speakers who are concerned with sounding gay-masculine as more likely to place importance on the indexicality of /s/ than those who do not voice concerns with sounding gay-masculine. This awareness spans both indexical meanings of /s/ (gay-masculinity and femininity), and these speakers thus take up the feminine meaning to protect themselves from cisgender listening judgements. I additionally discuss additional possible factors including transition stage. Reinterpreting my predictions as to concern with cisgender perceptions and relationship with hegemonic femininity, I find that speakers primarily seem to prioritize their safety, rather than disregarding cisgender judgements. I call for future research to look more closely at sociolinguistic resistance.

These conclusions emphasize the difficulty of trans speakers' linguistic position and the injustice of hegemonic norms, and the marginalizing influence of priviliged listening practices. I emphasize the power speakers have to protect themselves from these norms and the depth of speakers' indexical knowledge, while acknowledging that these efforts would not be necessary in a trans-safe world.

# Chapter 1

# Literature Review

## 1.1 Theoretical Motivations and Background

This thesis belongs to a lineage of research in the field of language and gender. Scholars of language and gender look to the gendered indexicalities of everyday linguistic behavior as a medium for gender performativity. For the bulk of this thesis, I follow the tendency of language and gender scholarship in centering indexicality as the primary framework of analysis. Although performativity is the point of departure for much of language and gender research – especially following the importance of performativity within gender studies after the foundational work of Judith Butler (1990) (see also Bucholtz 2014) – indexicality provides a number of theoretical tools, as reviewed below, which make for a more actionable lens of analysis (Henry and Hall 2025). Sociolinguists study the "social-indexical performativity of language-in-use" (Silverstein 2023, p. 17) considering performativity to be mediated by indexicality (Henry and Hall 2025). For this reason, I move forward by investigating indexicality. In this chapter, I first introduce recent approaches to indexicality in language and gender research, demonstrating the turn towards an approach which considers speaker and listener together, and underlining theoretical questions which still remain. I then review the language and gender literature on /s/, arguing that the existing scholarship is unable to provide complete answers to these theoretical questions.

Indexicality comes to sociolinguistics through semiotics, the study of meaning-making. In this thesis I adopt the model of semiotics put forth by Gal and Irvine (2019), who synthesize prior approaches, placing critical emphasis on the procedural aspect of meaning making by which phenomena come to be meaningful. In their model, individuals make meaning of their world through conjectures. Drawing upon their knowledge and understanding of the world, they recognize a phenomenon and conjecture that it may be a sign, i.e. it may be meaningful. They conjecture that the sign signals (means or represents) a particular object, its meaning. Essential to note here is that these conjectures constitute the sign-object relationship; nothing is a sign in itself, independent of a perceiver conjecturing that it is a sign and it signals an object (Gal and Irvine 2019, p. 89). Also important is

that these conjectures are based on the perceiver's knowledge and understanding of the world, and which are constructed and mediated by their cultures, communities, and individual social identities– their lived experience (Gal and Irvine 2019; see also Calder 2021). Part of a conjecture is its "grounds": the reasoning the conjecture has as to why its phenomenon is a sign and why it signals this object. A conjecture is considered to be 'indexical' if its grounds are the knowledge or belief that the sign often co-occurs with, points to, or is caused by its object (Gal and Irvine 2019, p. 96).

Sociolinguistics often focuses on finding correlations between differences in language use (eg. pronunciation, grammar, word choice) and differences in social identity, persona, or context (Eckert 2012). Particular linguistic forms are then said to index the social identities of those who use them, on the grounds of this relationship of co-occurrence (Eckert 2012). As discussed using the example of voice pitch above, if a speaker uses a particular linguistic variant, and the listener believes that form to be commonly used by those of a particular gender, they may make a conjecture on the grounds of this co-occurence, considering the use of that form to index that speaker's gender. That there are two actors in play here, a speaker and a listener, led Bucholtz and Hall (2005; 2004; 2010) to understand the indexical-performative construction of gender (and identity categories more broadly) as an intersubjective process.

Bucholtz and Hall characterize this intersubjective process as a negotiation between the indexical goals of the speaker and the indexical goals of the listener (eg. Bucholtz and Hall 2005, p. 606). Although it is possible that a listener may construe an indexical meaning from a linguistic form used by a speaker who has no knowledge of, or intent to make use of, that form's indexicality, agentivity or intentionality on the part of the speaker is common (whether conscious or unconscious). Speakers have interactional goals, and hope that their behavior will be construed as indexical of those goals, whether it be related to identity, eg. presenting feminine, or other aspects such as persona and stance, eg. seeming like a 'diva' (Podesva 2007). This agentive view arose principally in reaction to earlier sociolinguistic models which stripped speakers of agency and individuality, instead understanding their linguistic behavior simply as a product of speakers' static, essentialized identities (Eckert 2012). In practice, this turn reframes "she speaks this way because she is a woman" as "she chooses this form to index femininity." Acknowledging that individuals have choices about how to present themself and granting them authority over their self-identification is also a move towards alignment with queer and trans understandings of identity; trans people in the US often consider gender identity to be self-determined (see Zimman 2017b), rather than a consequence of bodily characteristics. Despite the value of emphasizing the speaker's goals to legitimize their identities and agency, it is nonetheless the case that the listener is the powerful other side of this negotiation.

Inoue (2006; 2003) argues that in cases where the listener occupies a privileged social position of power over the speaker, they may inhabit a privileged 'listening subject' position in which the beliefs required to uphold this social hierarchy may play a primary role in their construal of that speaker's indexical behavior and

override or erase the speaker's intentions. Studying the emergence of the notion of 'women's language' in Japan, Inoue finds that to Meiji period elite male intellectuals, schoolgirls inhabited a social position which represented a modernizing breach from traditional Japanese social structure, and a threat to their power. For this reason, when they overheard schoolgirls using the verb ending forms *teyo* and *dawa*, they construed these endings as indexical of degeneracy and social decline. Of course, schoolgirls themselves did not intend to use these forms to index a 'degenerate' identity; the purported degeneracy these forms were said to represent was a social fabrication rather than a truth about any actions these girls were involved with. In this way, male intellectual overhearers constructed a listening subject position which made negative judgmental conjectures about the indexicality of schoolgirls' speech, and constructed a fictional object for the these speech forms to signal. The male listening subject sets the narrative, erasing schoolgirl speakers' semiotic agency.

Following this important insight, research has extended theorization of the listening subject to consider race and queerness, researching white (Flores and Rosa 2015) and cisgender listening subjects (Calder 2021; Heffernan 2023; Jessee and Calder 2025; Calder and A. Steele 2025). Flores and Rosa (2015) find that when white listeners racialize speakers as non-white, they justify their position of power over those speakers by construing those speakers' speech as indexing an underspecified 'inappropriateness' for the situation (eg., the classroom), regardless of those speakers' adherence to linguistic norms for standard English, constructing a white-supremacist white listening subject which will always deem non-white speech deficient and indexical of a supposed non-white inferiority. Jessee and Calder (2025) find that cisgender listeners apply stricter gendered listening standards to an audio clip of a voice when they are told the speaker of the clip is a trans woman than when told the speaker is a cis woman, similarly situating sufficient gender performance out of reach for transgender speakers in a move to deny their identities.

Drawing on these concepts of white, cis, and male listening subjects, Calder (2021) and Calder and A. Steele (2025) raise an important critique of the sociolinguistic project, questioning which listening subjects researchers themselves may be inhabiting in their interpretation of speakers' language use. Following the conjecturing model of Gal and Irvine (2019), they note that researchers collect numerical correlation data between linguistic forms and social aspects, and use this as the knowledge on which to base their conjectures of indexical links. Incorporating the research on the listening subject, they conclude that although this data may seem objective, empirical, and scientific, purely describing the world as it is (see also Zimman 2021a), listening practices inevitably are biased by the sociocultural positioning of the researcher and the scholarship that researcher builds upon. This critique raises the question of how to research the social-indexical performativity of language-in-use without taking up and reinforcing violent listening practices and marginalizing speakers who are not white, cis, heterosexual, middle class men. Work which addresses this question employs advanced models of indexicality which encourage sociolinguistic study to move from simply asking what

meanings participants index to investigating the socio-cultural structure of individuals' indexical knowledge, and the intersubjective process of indexicality (Calder 2021; Silverstein 2003; Silverstein 2023; Barrett and Hall 2024). The concepts of the indexical field as adapted by Calder (2021), and indexical (mis)recognition as developed by Barrett and Hall (2024) have been used to model indexical knowledge and the relativity and intersubjectivity of indexicality.

The concept of the indexical field, introduced by Eckert (2008), has been adopted by Calder (2021) to capture the way in which a particular phenomenon may be conjectured to have a multiple meanings, both within a single group of people and across different groups of people. The concept of the indexical field emphasizes, connecting to Gal and Irvine's (2019) semiotic model, that indexical relationships are conjectured through listeners' knowledge. In this way, a particular linguistic form may be construed as a sign of any number of meanings, depending on which knowledge the listener uses to conjecture the relationship, and what conclusion they come to in conjecturing. Although this knowledge is mediated and constructed differently by each individual's social and cultural lived experience, it is nonetheless to a large extent shared. For example, Inoue's (2003) listening subject was not a single Meiji-era Japanese male intellectual, it was an ideological position constructed and inhabited by this male intellectual social class. Individuals who inhabit similar social and cultural backgrounds and social positions, as well as co-members of communities, are likely to draw upon a similar base of knowledge, and are likely to consider certain phenomena signs with a particular set of possible meanings. Calder (2021) conceptualizes the indexical field as this set of meanings which may be shared by a set of people with shared epistemology. By making explicit whose indexical field researchers are investigating and interpreting signs through, the fact that this interpretation is coming from particular perspectives is forefronted, preventing the universalization of the hegemonic indexical field.

While this makes progress towards a more critical and less biased move to interpreting signs, and much research has been productively done to investigate the indexical fields within particular local communities, especially community-specific systems of categorization and meaning, it is nonetheless untenable to simply split the world into non-overlapping spheres of shared knowledge and shared indexicality. Individuals bring their own indexical knowledge into interaction, and are likely aware of a number of indexical fields. The question that arises is: what happens when listeners apply an indexical field differently than a speaker intends them to, or when there is a mismatch in indexical knowledge?

Barrett and Hall (2024) model these situations using their concepts of the indexical inventory and indexical (mis)alignment/(mis)recognition. For them, an individual's indexical knowledge is structured in an indexical inventory, which is "an individual's full set of indexical fields." (p. 132) When individuals interact and engage in sociolinguistic meaning-making through shared indexical meanings, they are said to be in indexical alignment. In this case, if a speaker uses a linguistic form that they know to hold a particular indexical meaning within one of the indexical fields in their inventory, their listener would recognize (in Barrett and

Hall's term) their behavior as indexical of the meaning on which they are aligned, interpreting it in line with the speaker's intentions. However, in cases of indexical misalignment, the speaker's intended interpretation does not come through: the speaker employs a linguistic form with the intent to index a particular meaning in their inventory, but the listener does not interpret the form as indexing that meaning. Although it is possible the listener may simply be unaware of the indexicality the speaker intended to make use of, Barrett and Hall (2024) emphasize that misrecognition is often politically motivated, whether this motivation be resistance against, or reproduction of hegemonic power.

For the purposes of this thesis, it is most relevant to review the marginalizing side of misrecognition, by which misinterpretation of signs is a tool for the reproduction of power. Barrett and Hall (2024) raise the example of the transition of generic language in English from masculine (eg. *he/mankind*) forms to gender neutral forms (eg. *they/humankind*). They note that in this transition, feminists constructed a new indexical meaning around generic masculines, considering these forms an index of misogynistic views in those who use them. In reaction, anti-feminists deny this new indexicality, considering generic masculines grammatical fact and misrecognizing, or refusing to recognize, new meanings. Here, Barrett and Hall consider the anti-feminist approach to be an example of their concepts of indexical presumption – whereby those in power presume that their indexical inventory is universal and correct – and indexical denial – in which knowledge of indexicalities against one's interests are denied.

When considered together with the concept of listening subjects, indexical presumption and indexical denial emerge as tactics which privileged listening subject positions might employ to in interpreting meanings from linguistic behavior. Taken as a whole, the concepts of the indexical field, listening subjects, and tactics of indexical misrecognition do the work of, in Barrett and Hall's terms, relativising indexicality. This makes it no longer tenable for researchers of sociolinguistic indexicality to draw correlations and make their interpretations from there; instead, they must work to understand the procedure by which meaning arises and all of the factors which influence speakers' choices in language use. In particular, this thesis is interested in asking how marginalized speakers deal with these listening practices, hoping to contribute to an understanding of how speakers model listeners' indexical inventories and how they make use of these models in making (conscious or unconscious) linguistic choices.

Although all speakers may to some extent model their listeners' indexical inventories, I choose to focus on the case of marginalized speakers as a group who is under the most pressure to do so. Barrett and Hall (2024) note that indexical meanings are not evenly distributed, and I would add that the work of modeling listeners' indexical inventories is also not evenly distributed. When marginalized speakers interact outside of safe communities, they are at risk of being judged negatively by listening subjects which will presume the hegemonic indexical field and deny alternate indexicalities. Conversely, privileged speakers are unlikely to face any negative consequences due to ignorance of indexical fields belonging to marginalized groups; marginalized listeners do not have the power to inflict the

same violence as priviliged listeners might. In reaction to this situation of imbalance, marginalized speakers are coerced to protect themselves by learning hegemonic indexicalities and predicting how listeners may interpret their speech from an antagonistic listening subject position.

Heffernan (2023) reveals that transfeminine speakers are aware of the cisgender listening subject. Her study collected metalinguistic commentaries on transfeminine individuals relationships with their voices and experiences with being heard from an audio-recorded conversation she conducted between two US (South Asian American and White American) trans women and from anonymous transfeminine users on the Reddit online forum r/TransVoice. Commentaries see that cisgender listeners hear transfeminine voices through hegemonic norms and make binary judgements, categorizing voices as belonging to a man or a woman. Transfeminine speakers do not want to sound like men, and are aware of the possibility that cisgender listeners may make this judgements. In these commentaries, speakers additionally discuss what they do with this awareness and how they respond to this risk, a question I take as central to my study.

Heffernan finds that speakers simultaneously express feelings of pressure to modify their voices to avoid these negative judgements and feelings that they should not have to do so, exposing the conflict between "views of gender as self-determined, on the one hand, and of gender as intersubjectively constituted, on the other" (ii). These two approaches can be found in the commentaries from one of the audio-recording participants. This participant notes that when "interacting with the cis universe" she finds herself "deliberately modulating [her] voice" to make it "more girly," and that this is "a kind of survival tactic... to try and avoid any possibility that [she]'ll get misgendered" (23). In this way, the speaker acknowledges the risk of misgendering that comes from interacting with cisgender listeners, and her response is to change her language use in an attempt to align more closely with hegemonic feminine norms. However, she also asserts that "it's kind of just always incorrect to assume someone's gender," which and that she is uncertain about whether she would want to train herself to always use a more hegemonically feminine voice, as she wants to be able to "[think] about (.) [her] voice ... through uh- like the lens of other trans people", who would not apply the hegemonic indexical field, not assume her gender, and who would accept her voice as it is. This tension is around the decision to shift one's speech in the presence of a privileged listener who may inhabit a judgmental listening subject position is of primary interest to my study. Because she only investigated metalinguistic commentaries, Heffernan was not able to investigate how these decisions play out in transfeminine people's real interactions with cisgender listeners; this thesis aims to fill this gap.

Within the existing literature, I identify three sutdies which have considered changes in individual transgender speakers' language use based on context: Becker, Khan, and Zimman (2022), Gratton (2016), and Parnell-Mooney (2019). Becker, Khan, and Zimman (2022) find that trans men and nonbinary people on testosterone (assigned female at birth) use significantly less creaky voice (known popularly as 'vocal fry') in a reading-passage task compared to in a open-ended conver-

sation. As creaky voice is associated with femininity within the North American English hegemonic indexical field, they tentatively interpret these findings as evidence that in a more scrutinized form of speech (in which, I add, the notion of the cisgender listening subject may be more mentally present), these individuals purposefully move away from normative feminine speech in order to assert their identities as non-women.

This aligns with the findings in a study on a Toronto nonbinary friend group conducted by Gratton (2016). Gratton finds that variation between *-ing* and *in'* (eg. *working* vs. *workin'*) was more even in queer-safe spaces, where speakers used roughly the the same amount of both, and more differentiated in public, where speakers preferred one form over the other. In the hegemonic indexical field, *-ing* variant is hegemonically associated with femininity and *in'* with masculinity. Gratton finds that in public, speakers avoid the variant which is hegemonically indexical of their assigned gender at birth, understanding that cisgender overhearers are invested imposing that gender onto them and choosing to exaggerate their difference from that gender to avoid that perception. In queer-safe spaces, (in which, I add, the cisgender listening subject is not present), speakers were free to speak more neutrally and without regard for binary norms – Gratton notes that one speaker assigned male at birth mentions feeling comfortable making occasional use of a 'manly voice' in these spaces, as they know they will not face misgendering for doing so in the absence of cisgender listeners.

Parnell-Mooney (2019) also finds this sort differentiation, finding that in conversations with cis listeners, four transmasculine and two nonbinary speakers shift to pronounce /s/ further from the type of pronunciation expected of their assigned gender at birth. The trans men and the nonbinary person assigned female at birth shift further from hegemonically feminine /s/ pronunciations with with cisgender women listeners than when interacting with cisgender men, whereas the nonbinary person assigned male at birth shifts the opposite way. Although these studies present important empirical results, they do not integrate with the theoretical material discussed above – listening subjects, indexical fields, indexical recognition, and the modeling of a listener's indexical knowledge. They also do not include transfeminine speakers, and do not explicitly compare speech to a cis listener against speech to a trans listener. Filling this gap, I compare transfeminine speakers' indexical behavior across a conversation with a cisgender woman and a transgender woman with reference to their commentaries on their own gender, their voice, and being heard as a transfeminine speaker.

For my comparison, I focus specifically on the English /s/ sound due to the richness of existing research on its indexicalities and its use by queer and trans speakers, especially in connection to the sophisticated theories of indexicality discussed above. The remainder of this section introduces the literature on /s/ and reviews its connections to the concerns of transgender language and indexicality research.

## 1.2 The Sociophonetics of /s/

By varying the tongue position with which an /s/ is pronounced, a speaker can vary the frequencies (or pitches) of the sound produced; when the tongue position is fronter (closer to the teeth) the resulting /s/ has primarily higher frequencies, when the tongue position is backer (further from the teeth), the resulting /s/ has primarily lower frequencies (Fuchs and Toda 2010). A common measure of this frontness is the spectral center of gravity (CoG), which is the mean of all of the frequencies of an /s/ sound at a given moment in time, weighted for their amplitudes (volumes), measured in Hz. It is important to note at the outset that gendered differences in /s/ frontness do not arise from any anatomical differences associated with the sex binary, unlike other gender-correlated sites of variation such as overall vocal pitch; all people with typical vocal tracts have the anatomical equipment to produce roughly the same range of /s/ pronunciations (Zimman 2017c; 2021b).

Research on white, English-speaking, cisgender heterosexual men and women in North America – which did not include non-White, non-English-speaking, or queer speakers and listeners, a point to which I will return – find robust correlations between speaker gender and /s/ pronunciation. Flipsen et al. 1999 review 21 such studies, finding that women's /s/ pronunciations are consistently fronter than men's. Listeners in this population are aware of these correlations, and judge fronter /s/ to sound feminine, and backer /s/ to sound masculine. Strand (1999) finds that when asked to to categorize audio stimuli on a spectrum of /s/ frontness as either /s/ or /ʃ/[1] (/ʃ/ is similar to a very backed /s/), they place the boundary higher when told the speaker of the audio clip is a woman than when they are told it is a man, thus exhibiting a link between /s/ frontness and percieved gender. When listeners in this population percieve a mismatch in /s/ pronunciation and the percieved gender of a voice, they judge that voice to sound gay or lesbian. A number of studies have focused on this dynamic in perception of men's voices (eg. Levon 2014; Campbell-Kibler 2011). Munson, Jefferson, and McDonald (2006), replicates Strand's experiment, finding when listeners perceive a speaker to be lesbian sounding, they place the boundary between /s/ or /ʃ/ lower, thus expecting lesbians to back /s/.

Calder (2021) points out that these earlier correlational studies often interpret indexicalities, but fail to situate these indexicalities as specific to the populations in which they are found. It is insufficient to claim that fronter /s/ pronunciations index femininity and gayness and backer /s/ pronunciations index masculinity and lesbianness without specifying for whom these indexicalities hold. These indexical links are made specifically by cisgender heterosexual white listeners, who take the behavior of their own population as the norm, and interpret others through theis norm (an act of indexical presumption). Calder (2021) thus considers indexical links from /s/ frontness to femininity and gayness to be gender norms within the hegemonic indexical field for /s/, rather than neutral and universalizable patterns.

Calder (2021) proceeds to discuss two examples of research on outside of this

[1] /ʃ/ represents the English 'sh' sound, as in the beginning of 'shirt' or the middle of 'station'

hegemonic indexical field, one investigating Black American speakers' /s/ pronunciations, and one investigating queer nonbinary drag queens'. In the first of these studies, J. Calder and King (2022) measure the speech of find that cisgender heterosexual Black speakers in Bakersfield, California /s/ pronunciations were not differentiated by speakers' gender. They highlight that white normativity does not apply here; the men and women who participated in this study considered themselves normatively gendered cisgender heterosexuals, and cannot be said to be improperly indexing their genders, or indexing queerness. Within the indexical field of speakers in this community, Calder and King argue /s/ comes to be an index of white femininity/masculinity, and thus irrelevant to (or even to be avoided by) black speakers.

Looking at racialized aspects of /s/ in perception, Maegaard and Pharao (2015) and Pharao et al. (2014) present a similar finding in a different context. In Copenhagen Dutch, /s/ has been shown to pattern and be percieved similarly to the North American white cisgender heterosexual English-speaking population. Maegaard and Pharao find that young listeners (both white and nonwhite) in only hear fronted /s/ as feminine and gay when it is within the context of a prosodic pattern associated with middle-class whiteness, 'modern' standard Copenhagen Danish speech; in the context of a prosodic pattern associated with lower-class and immigrants, 'street' Danish, listeners instead hear fronted /s/ as associated with 'gangster'-ness and masculinity. Although Maegaard and Pharao do not apply the concept of Indexical fields to their work, it can be said that /s/ has two indexical potentials within the indexical field of these listeners, and that other indexical behavior, in this case the prosodic pattern, determines which set of indexicalities speakers take up – if the speaker sounds white with the 'modern' prosodic pattern, /s/ is interpreted through white gender norms, if the speaker sounds nonwhite with the 'street' prosodic pattern, /s/ is intepreted differently.

From these examples, two points are clear. First, local or marginalized communities' indexical fields are not simply independent of hegemonic meanings, but are instead interlinked with them, qualifying and reinterpreting hegemonic meanings. Second, variables like /s/ do not work on their own, but only come to be interpretable through a broader semiotic context; listeners in Maegaard and Pharao's studies could only interpret /s/ through its combination with prosodic types. In the second of the studies mentioned above, Calder (2019a) applies the theories of indexical orders developed by Silverstein (2023; 2003), and fractal recursivity developed by Irvine and Gal (2000) and Gal and Irvine (1995) in order to investigate this process.

As linguistic forms are construed as indexical signs and gain meaning within a particular indexical field, it is possible that new meanings will arise which build off of and are informed by the original meaning. This can take place both within a single indexical field, and cross-cutting the overlaps of multiple. The development of an indexical link between a linguistic form and a social meaning always already emerges informed by an uncountable chain of previous indexicalities and meaning-making, and for this reason is said to exist at the $n$th indexical order (Silverstein 2003). In the case of /s/, Calder understands the indexical link between the gender

binary and /s/ frontness to exist at the $n$th order within the hegemonic indexical field. When cisgender heterosexual white listeners' perception of speakers' gender does not match with their perception of that speaker's /s/ pronunciation, this recontextualization of /s/ usage forms the $n+1$st indexical order, in which a new interpretation must emerge. Here, the meaning of /s/ comes to be qualified by other construals of meaning, in that a listener's perception of a speaker's gender must be based upon other indexical conjecturing. That cisgender heterosexual white listeners construe this percieved mismatch in speaker gender and /s/ pronunciation as as gay/lesbianness evidences an erasure of transgender, nonbinary, and other queer identities; gender is assumed to be fixed, and all gender nonconformance is assumed to be gay or lesbian. Without these cisgender assumptions, the $n+1$st indexical meaning of a mismatch of percieved speaker gender and /s/ would simply be gender nonconformance, or perhaps that the speaker's gender was percieved incorrectly. With these assumptions, it comes to be gay/lesbianness. It is important to note that although the $n$th and $n+1$st indexicalities of /s/ are situated at different orders, they nonetheless inform and construct each other, and do not form a chronological order despite their ordered structure.

Within the San Francisco drag communities Calder (2019a) studies, an additional indexical order of /s/ is forged at the $n+1+1$nd level which is local to these communities' indexical fields, escaping the hegemonic field. A drag community Calder studies in the South of Market (SoMa) neighborhood of San Fransisco figures itself as more radically queer than another drag community in the Castro neighborhood. This is reflected by the differences in identities between these communities; SoMa queens generally identify as beyond the gender binary, many of them nonbinary or transgender, whereas the queens of the Castro neighborhood primarily identify as cisgender gay men. Calder finds that SoMa queens front /s/ more than Castro queens, and that this forges a new local indexicality to 'fierceness', anti-normativity, and and in-group status at the at the $n+1+1$nd level. Their diagram of this structure of meanings is included in figure 1.1. This emerges as an example of indexical orders cross-cutting indexical fields, where meanings in the hegemonic cisgender heterosexual white indexical field came to inform the creation of new meanings in the local, alternative indexical field.

As mentioned above, within Calder's model fronting /s/ emerges as deviant from gender norms only in those whose gender is understood as 'male' within the cisnormative conception of gender as bodily characteristics. It is perhaps surprising, then, that nonbinary and transfeminine speakers in the SoMa drag community, who would not like to be percieved as men, front /s/ exactly for the purpose of gender defiance, rather than gender conformance. It is helpful here to introduce a theoretical tool Calder employs to understand the interlinking of identity categories at these different levels of sex at birth, gender identity, sexuality, and trans/queerness– the semiotic process of fractal recursivity, as theorized by Irvine and Gal (2000) and Gal and Irvine (1995).

In fractal recursitivity, a semiotic opposition on one level becomes nested into itself, and this process can happen any number of times (Irvine and Gal 2000; Gal and Irvine 1995). In moving from the $n$th indexical order of /s/ to the $n+1$st (both

of which are within the hegemonic indexical field), the gender binary is nested into itself, creating a metaphorical structure where men and women are split into binary gender oppositions. This can be understood as distinguishing some notion of 'men who are men' from 'men who are women', and 'women who are women' from 'women who are men'. These become the metaphorical representation of gay men and lesbian women, since members of these groups are still fundamentally understood as types of men and types of women, relying on the aforementioned erasure of trans people and those beyond the binary. Is it then the case that by situating the distinction between cisgender gay Castro queens and nonbinary and transgender queer SoMa queens as nesting into the 'men who are women' branch, Calder as the analyst, and possibly the SoMa queens themselves, tacitly consider these queens as nonetheless a 'type of man/male'? I argue it is not. Instead, in my reading of Calder's work, SoMa queens reinterpret at the $n + 1 + 1$st level such that the fractal-recursive binary structure is broken; instead, the opposition is cisgender gay men vs. non-cisgender queer people, or as Calder articulates it, gay vs. queer. This binary still fits with the base level of /s/ indexicality, which is always present; it is not difficult for SoMa queens to figure gay men as less feminine and less subversive than queer non-cisgender people. From here, the indexicality of /s/ frontness as subversive queer non-cis femininity can be less problematically interpreted.

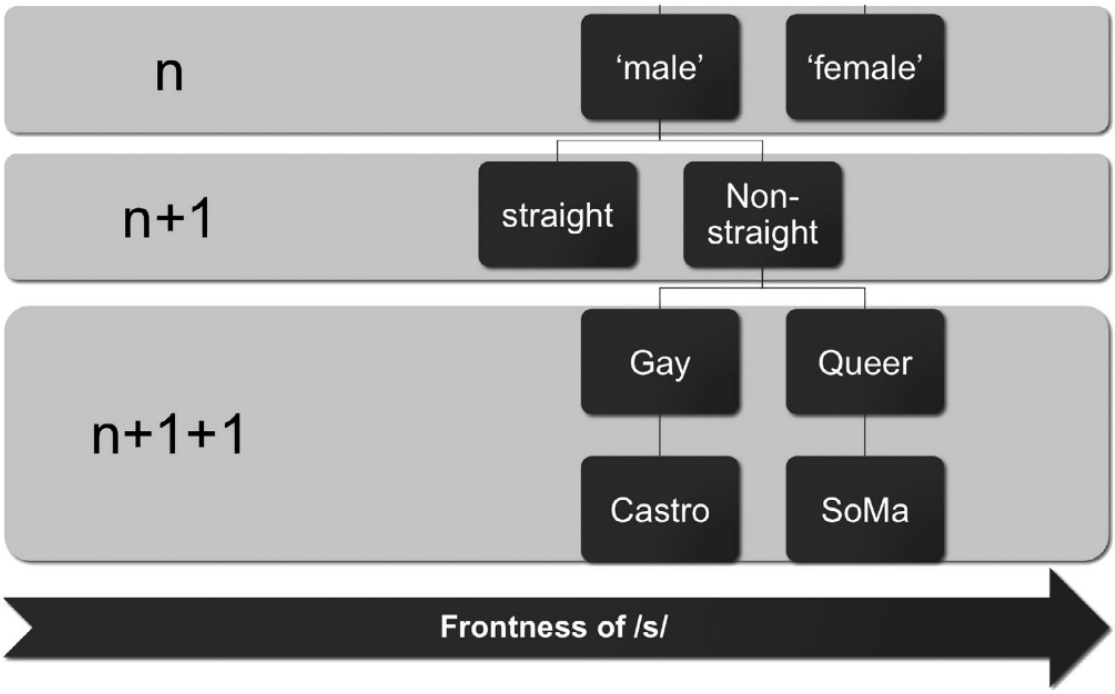



Figure 1.1: Diagram from Calder (2019a, p. 340) illustrating indexical orders of /s/ frontness and corresponding fractal recursive social oppositions across two San Fransisco drag communities.

A. J. Steele (2022) provides another example of /s/ being reinterpreted by those

outside of the gender binary. They investigate a racially-diverse community of Columbus, Ohio nonbinary individuals, finding that for Black speakers, fronted /s/ seemed to have the opposite indexical meaning, correlating positively with speakers' self-rating of their masculinity. For white speakers in this community, there was no correlation between /s/ pronunciations and masculinity ratings. This leads Steele to conclude that Black speakers subvert /s/ meanings not only to create a nonbinary identity and differentiate themselvelves from a cisgender norm, but also to repurpose and subvert a white norm. For white speakers, this dimension is not present, so the behavior is different. These patterns suggest that nonbinary speakers in this community forge further indexical orders which allow them to negotiate their specific racial and gender positions against white and cisgender normativities.

The studies discussed above illustrate the ability of speakers outside of cisgender heterosexual white populations to construct alternative and subversive meaning structures for /s/ which allow an interpretation of their unique speech practices as innovative and subversive, rather than inadequate and deviant. However, due to the dominance of the hegemonic indexical field, and the practicalities of social life, it is nonetheless the case that marginalized speakers outside of cisgender heterosexual white populations have to deal with these norms. Although it is important to note that these all speakers are socialized to believe in these norms, even if these norms will marginalize them, I focus on the most certain case, in which marginalized speakers interact with those these norms directly center and privilige.

That the hegemonic structure of indexicality for /s/ particularly excludes trans speakers is attested to by the findings of Jessee and Calder (2025). They investigate cisgender perceptions of transgender /s/ by replicating Strand (1999), comparing the placement of the boundary between /s/ and /ʃ/ when the perceiver is told the speaker is trans vs. not told they are trans (and thus is likely to assume that they are cis). They find that unlike in Munson, Jefferson, and McDonald (2006), where cisgender heterosexual speakers expect cisgender gay men and lesbians not to conform with the hegemonically normative /s/ pronunciation for their gender, transgender speakers are expected to hyper-conform– trans women are expected to front /s/ more than cis women by all listeners, and listeners who do not know a trans person also expect trans men to back /s/ more than cis men. Although indexicalities certainly would have changed in the 20 years between these two studies, it is nonetheless surprising that cisgender heterosexual white listener holds different expectations and perceptions for trans people than for cisgender gay men and lesbians, especially since it is counter to trans speakers' actual behavior; Podesva and Van Hofwegen (2015), Hazenberg (2012), and Zimman (2021b) find that trans speakers' /s/ pronunciations tend to be closer to the center of the spectrum, not as strongly gender-differentiated than cis speakers'.

Jessee and Calder (2025) analyze this through the lens of the cisgender listening subject, seeing that rather than basing perceptions on transgender speakers' actual speech, "the cisgender listening subject, which expects voices to conform to normative standards of gender appropriateness, places even stronger standards of

gender normativity on trans speakers than it does on cis speakers." (9-10) Here, the cisgender listening subject will only accept an idealized highly-conforming trans person, and construes real trans speakers as inadequately performing their genders. Because we do not live up to this idealized standard, cis percievers will misgender trans speakers, inevitably hearing them as the gender they were assigned at birth, rather than their actual gender. In combination with the $n + 1$st order of meanings for /s/, this means that attempts to move towards fulfillment of hegemonic gendered /s/ pronunciations are likely to be misrecognized as indexical of gay and lesbianness, rather than transfemininity and transmasculinity respectively. Trans speakers are 'damned if they do, damned if they don't' – if a trans woman does not front /s/, her transness may be questioned, and she may be heard as a man; if she fronts /s/, she may be construed only as a gay man, not a trans woman. In this difficult indexical situation, in which a trans speaker knows their cisgender listener may subject them to this hegemonic indexical field which excludes them, there are two primary options; defend oneself by moving towards adherence to these norms, or refuse to do so, resisting these norms and risking negative judgements. As mentioned above, transfeminine speakers studied by Heffernan (2023) consider both of these options, acknowledging the safety and possible gender-affirmation in taking up hegemonically normative femininity, while simultaneously not wishing to judge themselves by cisgender standards. Drawing upon all of this prior work on indexicalities of /s/, this thesis investigates whether transfeminine speakers shift towards hegemonically-feminine fronter /s/ pronunciations with cis listeners, and what motivations might determine the choice to shift or not shift in view

That individuals may change their /s/ pronunciations within interaction for gender-presentational and interactional purposes has been attested by Benesch (2022), Mendoza (2024), Saigusa (2016), Holmes-Elliott and Levon (2017), Levon and Holmes-Elliott (2013), Parnell-Mooney (2019), Calder (2019b), and Zimman (2017c). Holmes-Elliott and Levon (2017) and Levon and Holmes-Elliott (2013), in a study of within-speaker variation in /s/ pronunciation in British reality television shows, find that in one lower-class show, cisgender heterosexual women use backer /s/ in conversation with other women, whereas in a higher-class show, cisgender heterosexual women use fronter /s/ in this context. In order to explain these contradictory results, they conclude that the interlocutor's gender may not be the best predictor, as in both shows women back /s/ in speech activities considered more threatening. As discussed below in the methods chapter, the recorded conversations I conducted attempted to control for this stance-based shifting through the structure of two collaborative-valenced speech activities.

Moving to investigate /s/ shifting in queer speakers, Calder (2019b) finds that as SoMa drag queens get into their drag, visually transforming into their drag persona, their /s/ pronunciations get progressively fronter to index their change in persona. Benesch (2022) finds that among genderfluid people, who experience different genders at different times, shifts in /s/ index these shifts in gender, however they do not necessarily correlate with typical scales of gender such as masculinity vs. femininity, but rather with personal, alternative scales such as 'flannel' vs. 'glitter'. Mendoza (2024) presents a case study of a drag queen in Wynwood, Mi-

ami, who uses fronter /s/ pronunciations when discussing the aspects of their drag taken from white, higher-class femininity, but uses backer /s/ when discussing to their more anti-normative *chonga* drag persona. *chonga* is a queer subversive Latinx femininity which may look like "unshaved body hair and beards, as well as hoop earrings and dramatic lip liner" (307). The polyvalence of this queen's drag persona is indexed through these changes in /s/. In a similar way to the findings of Calder (2019b) and A. J. Steele (2022), this queen reinterprets the indexicality of fronted /s/, using it not to index of femininity in general, but rather to index a specific drag satire of white cisgender heterosexual higher-class femininity. In this way, meanings evolve outside of the hegemonic field.

I choose to focus on /s/ for my study, rather than other linguistic forms bearing social indexicalities, because of the theoretical depth of the research that has already been conducted on /s/; as reviewed above, this research has captured a nuanced and detailed view of the hegemonic indexical field of /s/, how it is marginalizing, and how those who are marginalized by it reinterpret and construct its meanings subversively within their communities. However, in the /s/ literature, as well as in sociolinguistic research more broadly, a gap I address is the dearth of research looking specifically at situations of indexical constraint, analyzing how marginalized speakers navigate listeners' indexical fields which marginalize them. Comparing transfeminine individuals' /s/ pronunciations in conversations with another transfem, where they can adopt their own indexicalities more freely, against their /s/ pronunciations in conversations with a cis woman, is illuminating here. More broadly, this research also addresses the absence of within-speaker research on transfeminine speakers– although there have been of studies of within-speaker variation in transmasculine people (eg. Zimman 2017c; Parnell-Mooney 2019), nonbinary people (Gratton 2016), and drag queens (Calder 2019b; Mendoza 2024), there have been no such studies focused on how transfeminine speakers speak differently in differing situations. Of specific interest is the question of individuals' navigation of the $n+1$st indexical order of /s/ frontness linked to gay masculinity; whether transfeminine speakers understand this to be a possible judgement of the cisgender listening subject, and whether this is taken into account in the choice to shift /s/. In sum, this thesis pursues the research question:

> Do transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener? What factors determine this choice?

Based on the literature reviewed, I predict:

1. Speakers who express greater feelings of danger, discomfort, dysphoria, or desire to pass around cisgender people are more likely to shift;
2. Speakers who differentiate themselves from or hold attitudes of resistance against hegemonic femininity are less likely to shift;
3. and Speakers who express concern with gay-male misrecognition are less likely to shift.

These predictions are based on the literature reviewed above, primarily the structure of the indexicality /s/ within the hegemonic indexical field it has established. Because of transfeminine people's awareness of the cisgender listening subject's application of hegemonic norms, I expect that transfeminine speakers' (conscious or unconscious) decision making around /s/ shifting will be based upon their relationship with the meanings of /s/ in the hegemonic field, as well as their relationship with hegemonic feminine norms itself; if speakers wish to distance themselves from hegemonic femininity, they will be more likely to disregard the judgements of the cisgender listening subject and refuse to take up a fronter /s/ with a cisgender listener. However, if speakers feel more threatened than defiant, feeling greater pressure to conform to hegemonic femininity, they are likely to shift towards fronter /s/ pronunciations in self-defense and to match up to the hegemonic standards. Yet, if speakers are concerned with being mistaken for a gay man via their voice, they may feel that even if they would be able to benefit from indexing hegemonic femininity, that /s/ is not an available resource through which to do so.

# Chapter 2

# Methodology

In order to investigate whether and how transfeminine individuals change their pronunciations of /s/ based on the trans status of their conversation partner, and to be able to interpret the possible motivations behind these patterns of change, I recruited four transfeminine individuals to be speakers and two women, one cisgender and one transgender, to be conversation partners (CPs). Each speaker participated in an audio-recorded, semi-structured casual conversation session with the each of the conversation partners, then completed a demographic survey, then participated in an audio-recorded semi-structured interview with me about their transgender experience and their relationship with their voice. This order of study activities was selected to minimize priming effects, since it is possible questions asked in the survey and interview could effect subsequent linguistic behavior.

## 2.1 Participants

All participants were students of Reed College between the ages of 18 and 25, recruited by flyers, word-of-mouth, and emails to the Reed College student body mailing list and to individual friends-of-friends. All speakers identified as transfeminine, and both conversation partners identified as women; one identified as cisgender and one identified as transgender. Speakers are referred to by animal-name pseudonyms, and the conversation partners are referred to as 'the Cis CP' and 'the Trans CP' respectively.

### 2.1.1 Survey

On the survey, participants were asked to provide any and all gender, sexuality, and race labels they identify with, as well as their pronouns. With respect to gender identity, I additionally asked participants to rank on the survey how much they generally identify with femininity and masculinity on scales of 0 to 100. However, I decided not to make use of these rankings for my analysis, as most of the participants seemed to find them uncomfortable and difficult to answer meaningfully. For the participants who were not the Cis CP, the survey also asked whether the

participant was currently on hormone replacement therapy (HRT) and for how long they had been on it, whether they had received voice feminization therapy and whether they had received voice feminization surgery. All of the trans participants were on HRT, and no participants had received voice feminization therapy or voice feminization surgery. Questions were also included to understand speakers' linguistic backgrounds– what languages they speak, whether they have primarily spoken English throughout their life or have had periods of primarily speaking other languages, the region(s) in which they grew up, and whether they had received non-gender-related speech therapy. Survey questions are provided in appendix A.

### 2.1.2 Interviews

After conversation sessions and surveys were completed, I also conducted 20-30 minute audio-recorded semi-structured interviews with the speakers. As a starting point, I asked questions intended to elicit background on each participant's individual relationship and history with gender and sexuality beyond the labels collected in the survey. From here, I hoped to see how participants position themselves with respect to hegemonic femininity, and how this relationship may have changed and developed over the course of transition. The discussion that followed from these questions tended also to address the question of 'outness'; what this meant for each speaker, and how private their trans and queer status was. After this, I moved to asking questions about participants' relationships with their voices. Some of these questions were modeled after those used by Heffernan (2023) to investigate speakers' understandings of the cisgender listening subject, asking speakers to talk about how they feel their voice is perceived by cis listeners and whether they might speak differently around different people. I did not directly ask speakers whether they were concerned that their voice would be mistaken as a gay man's voice, instead allowing the topic to naturally arise if it was salient to the participant. I also included in this portion questions about sexuality and the voice, such as "Do you feel that your voice sounds lesbian?", in order to get a sense of the relationship sexuality may play in determining /s/ pronunciation and shifting behavior. A full list of questions is provided in appendix B. Participants' demographic information and gender background from survey responses and interviews are summarized in tables 2.1.2 and 2.1.2.

## 2.2 Conversations

Before the survey and interviews, speakers participated in two 20-30 minute audio-recorded conversation sessions, one with each of the conversation partners. As these recordings would be subject to acoustic analysis of /s/, they were conducted in a sound-isolated recording booth using clip-on lavalier microphones (on the speaker as well as the conversation partner) connected to a Zoom h4n recorder. Participants were told to spend the first 10 minutes of the recording time collabo-

| Participant | Gender | Sexuality | Race | First Language |
|---|---|---|---|---|
| Mantis (she/her) | (Trans) Woman, (Trans) Girl | Queer (Prefers to date Trans men and Nonbinary People) | Asian American, Taiwanese, Japanese, Wasian | English |
| Alpaca (she/it) | Woman, Girl | Lesbian | White | English |
| Squirrel (she/her) | Transfem, Trans Girl | Bisexual (Prefers to date Fem People) | White | English |
| Jellyfish (she/her) | Woman | Lesbian | White | English |
| Trans CP (she/her) | Woman | Ace spectrum, Demisexual | East Asian, Han Chinese | Mandarin |
| Cis CP (she/her) | Woman | Lesbian | white | English |

Table 2.1: Participants' Demographic Identity Information, from survey and interviews

ratively playing a word game of their choice (crossword, New York Times Wordle, New York Times Connections, etc.). This prompting was intended to get participants comfortable talking with each other on the recording and to take up a collaborative and friendly stance towards one another. Controlling for stance in this way was intended to isolate stance-related variation in /s/, as indexicalities between /s/ and confrontational/threatening stances have been found (see Holmes-Elliott and Levon (2017)). Speakers and CPs were asked to time themselves and stop playing the game after the first 10 minutes of the recording, and to spend the rest of the time of the recording, another 15-20 minutes, getting to know each other, trying to find things in common, and talking about shared experiences. Since Alpaca and the Trans CP already knew each other, they were asked to catch up and chat as they would normally. The rest of the speakers and conversation partners did not know each other. The second portion of the conversation was intended to elicit free-flowing casual talk while maintaining similarly friendly and comfortable stance orientation. Although there were some moments of awkwardness, conversations did not include disagreements, arguing, or other confrontational elements and primarily maintained a lighthearted and friendly tone.

| Participant | Time Out | Time on HRT |
|---|---|---|
| Mantis | ∼3yr | 1y8mo |
| Alpaca | ∼4yr | 3yr9mo |
| Squirrel | 9mo (primarily online) | 3mo |
| Jellyfish | >3yr | ∼2yr |
| Trans CP | 4yr | 3yr |

Table 2.2: Participants' Transgender Background Information, from survey and interviews

## 2.3 Analytical Approach

In order to answer the first portion of my research question – whether transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener – I take a quantitative empirical approach. I operationalize spectral center of gravity (CoG) of /s/, measured in Hz, as the dependent variable, and conversation partner (Cis CP vs. Trans CP) as the independent variable. In order to measure CoG of /s/, I produced a time-aligned plain-English transcript of each recording. As a first pass, I used Deepgram's Nova 3 speech recognition model, specifying that it should transcribe utterances (rather than paragraphs or as one block of text) and annotate each utterance with who it was spoken by: the speaker or the conversation partner. I then used a Python script to convert the output of this model into a time-aligned transcription file format (TextGrid) usable by the linguistic acoustic analysis software Praat (Boersma and Weenik 2023). I then manually corrected the transcripts in Praat to ensure accuracy. I then used the Montreal Forced Aligner with the included English acoustic model and dictionary to automate the production of a time-aligned phonetic transcript, which would specify, for both the speaker and the conversation partner, the start and end time in the recording of each /s/ pronunciation (McAuliffe et al. 2017). From there, I used DiCanio's (2021) Praat script to measure the CoG of each of these /s/ pronunciations in each of the recordings for the speakers and conversation partners. It was important to measure not only the /s/ pronunciations produced by the speaker participants, but also those produced by the conversation partner participants, in order to determine the extent to which the speakers may have been accomodating to the /s/ pronunciations of the conversation partners (and vice versa). I used the statistical analysis software R (R Core Team 2025) to combine, statistically analyze, and visualize the data.

In determining the second half of my research question – what factors determine whether speakers shift to a fronter /s/ pronunciation with the Cis CP – I take a qualitative approach. I use interview responses to qualitatively assess the extent to which each speaker aligns with the three criteria in my predictions, quoting and discussing instances in which speakers:

1. express feelings of danger, discomfort, dysphoria, or desire to pass around cisgender people;

2. differentiate themselves from or hold attitudes of resistance against hegemonic femininity;

3. and express concern with gay-male misrecognition.

This approach is intended to encourage my analysis of speakers on their own terms, rather than simply comparing them against a white cisgender heteronormative standard which will inevitably find them somehow faulty (see Calder and A. Steele 2025; Zimman 2021a). Instead, I aim to understand the connections between speakers understandings of their voices and their vocal behavior, underlining instances of strategic deployment of semiotic resources, as well as the oppressive limitations imposed by cisnormativity.

# Chapter 3

# Results

| | Number of /s/ tokens | | Duration of Recording | |
|---|---|---|---|---|
| Speaker | w/ Trans CP | w/ Cis CP | w/ Trans CP | w/ Cis CP |
| Mantis | 177 | 143 | 21m48s | 14m31s |
| Alpaca | 476 | 244 | 30m56s | 21m53s |
| Squirrel | 388 | 401 | 25m04s | 29m49s |
| Jellyfish | 444 | 283 | 26m27s | 20m08s |

Table 3.1: Table of number of /s/ tokens measured and duration of each recording.

| | Mean /s/ CoG (Hz) | | | |
|---|---|---|---|---|
| Participant | Overall | w/ Trans CP | w/ Cis CP | $p$-value |
| Mantis | 7231 | 6638 | 7964 | $<0.001$ |
| Alpaca | 5758 | 5899 | 5483 | $<0.001$ |
| Squirrel | 6357 | 6345 | 6369 | $0.8263$ |
| Jellyfish | 6154 | 6039 | 6334 | $<0.01$ |
| Trans CP | 5133 | N/A | N/A | N/A |
| Cis CP | 6835 | N/A | N/A | N/A |

Table 3.2: Table of Mean /s/ CoG used by each speaker, overall as well as separated by conversation partner (CP). $p$-value indicates the likelihood that the difference in means between Trans and Cis conversation partner could be a result of random chance; $p$-values of $<0.01$ indicate a statistically significant difference in means.

Table 3.1 lists the duration of each recording and the number of /s/ tokens produced by the speaker participant in each recording. Table 3.2 and figure 3.1 present an overview of participants' /s/ pronunciations. Zimman (2017c) reports that so-called 'reference' studies on white English speakers presumed to be cisgender and heterosexual report a 'typical' /s/ CoG range of 6,400Hz–8,500Hz for women and 4,000–7,000Hz for men. That only Mantis and the Cis CP fell inside

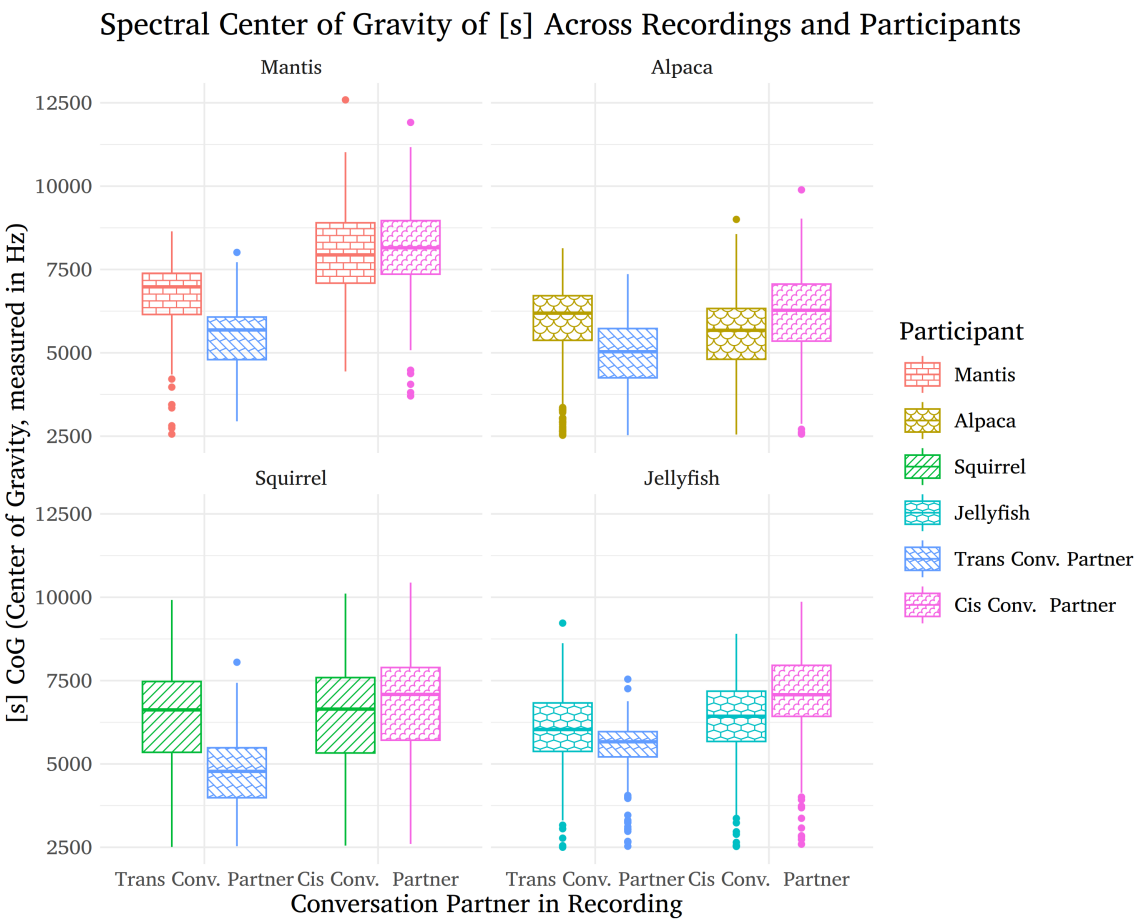



Figure 3.1: CoG of /s/ pronunciations in each recording, by each speaker.

the range expected of women, and that only Mantis fell outside the range expected of men, despite the fact that all of my participants identified as women or girls, points towards the inadequacy of these 'reference' values outside the cisgender heterosexual white population (recall that the Cis CP is a lesbian). I argue that this discrepancy should not be read as an indication of inaccurate or flawed measurement methodology, however, as my speakers' means do align with previous measurements for queer and trans women: the white trans women and cis lesbians Podesva and Van Hofwegen (2015) study in Redding, California exhibit mean /s/ CoG 6,000Hz and 6,350Hz respectively, and the white trans women Hazenberg (2012) studies in Ottawa, Ontario exhibit a mean /s/ CoG of 6,600Hz. These measures align much more closely with my speakers, with the exception of the Trans CP and Mantis, who occupy the extremes of my data. I discuss these individual cases below, but to point here is that I interpret the general correspondence of my data with previous studies of trans and queer women as confirming the validity of my measurements.

With respect to shifting, three of my four speakers, Mantis, Alpaca, and Jellyfish, exhibit statistically significant shifts in /s/ CoG from the recording with the

Trans CP to the recording with the Cis CP, and one speaker, Squirrel, did not. Of the three speakers who shifted, two, Mantis and Jellyfish, shifted in the direction of fronter /s/ pronunciations producing higher CoGs when talking to the Cis CP. This accomodation shifts their speech into closer alignment with hegemonic expectations of the feminine voice. The other speaker who shifted, Alpaca, shifted in the opposite direction, shifting to backer /s/ pronunciations with the Cis CP, moving away from the hegemonic norm in this context. With respect to the degree of shifting, Mantis exhibited the largest shift with a difference in means of $>1000$Hz, while Alpaca and Jellyfish shifted more subtly. Squirrel, who did not shift, maintained the same (relatively fronted) /s/ pronunciation across both conversations with a mean CoG of 6357 Hz.

In the sections that follow, I consider each speaker individually, qualitatively assessing their commentaries in the interview for the attitudinal factors listed above in section 2.3 in reference to their /s/ shifting behavior. I attempt to explain speakers' /s/ pronunciations in comparison to prior literature, my understanding of their identity and background, and these attitudinal factors. Following this, chapter 4 consists of a synthesis of these analyses, and a discussion of the implications of this data for the theories and research questions presented in chapter 1.

## 3.1 Speaker 1: Mantis

Figure 3.2 zooms in on Mantis' /s/ pronunciations. Her /s/ behavior is distinct from the other speakers' – her shift is the largest, and her overall mean CoG is the highest. As mentioned above, her /s/ pronunciations fall within the hegemonically-expected 'reference' range for cisgender women with an overall mean of 7231hz. In my interview with her, Mantis told me that she has "had so many people come and tell [her] that they thought [she] was a cis woman with [her] voice"; it is possible that her /s/ pitch contributes to these perceptions. She notes that these instances of having a cisgender-passing voice are the reason that she "used to worry a lot about [her] voice, but now it doesn't bother [her] as much." Here, she constructs a narrative about the development of gender confidence over the course of transition. Alpaca, Jellyfish, and the Trans CP – the other participants in the study who had been out as trans women for multiple years – all related similar stories of development in my interviews with them. Although this may seem like an expected part of a gender transition, it is nonetheless important to note, because for my speakers, part of these narratives was the development of a greater comfort with deviation from hegemonically-normative standards of femininity. Mantis notes:

> "I don't necessarily consider myself to be a woman in a way that a binary woman is a woman, but I also don't say I'm nonbinary, because I very much am like 100% woman, it's just, not in anybody else's way."

These narratives strike me as constructing a stance of resistance against the norms of unitary hegemonic femininity; in understanding her womanhood as unique to

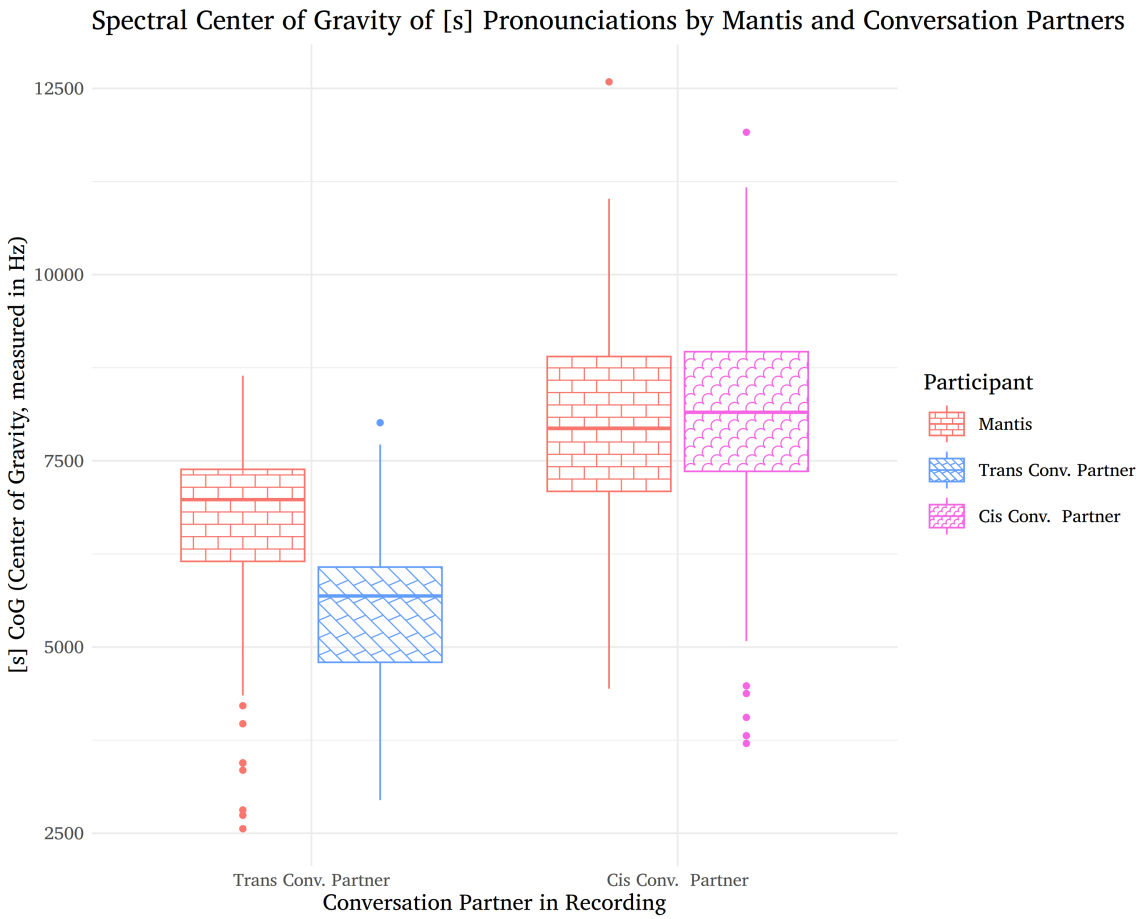



Figure 3.2: CoG of /s/ pronunciations in each recording containing Mantis, her pronunciations and the pronunciations of the conversation partners.

herself, Mantis asserts that she does not have to adhere to 'anybody else's way' in order to be '100% woman'. Here, Mantis emerges as a counter-example to one of my predictions; she shifts, despite distancing herself from cisgender norms.

An additional counter-example provided by Mantis relates to the issue of misrecognition. She does attest that sounding like a gay man is a concern, but this does not lead her to avoid fronter /s/ pronunciations with the cisgender listener as I predicted. When asked how her experience with her own voice has developed over time, she began her answer by noting that she has "always had like 'gay voice' or like feminine voice". When I asked whether her voice dysphoria was primarily the concern of not sounding feminine enough, she replied:

> "... it was, more specifically, like 'oh I just sound like a gay man'. And then there would be gay men who have gay voice ... and I'd be like 'oh I can't distinguish my voice from theirs', and it's just very frustrating, and I don't want people to think like that of me."

From this answer, it was clear that concerns about misrecognition represent an important part of Mantis's relationship with her voice. I discuss the possible in-

terpretations of this apparent mismatch in attitude and behavior in the following chapter. At this point is is sufficient to note that Mantis stands here as a counterexample to my prediction that speakers concerned with misrecognition as a gay man would avoid fronter /s/ pronunciations with the Cis CP.

On the other hand, Mantis did provide evidence to support my first prediction ,that speakers who reported greater feelings of pressure to conform or 'pass' with their voice in the presence of cisgender listeners would be more likely to shift to a fronter /s/ pronunciation with the Cis CP. In our discussion of voice dysphoria, Mantis mentions:

> "in conference sometimes ... I'm like, I don't know, maybe I shouldn't speak, because the person who just spoke before me just sounded like, so, woman, and now if I speak, people are gonna think I sound like a man. ... especially with cishet[1] women ... it's almost like a competition."

In these moments, Mantis feels that cisgender women's speech becomes the standard for feminine speech, and that the differences between her voice and the voices of cis women around her mark her speech as deficient and male. In order for feminine speech to be a 'competition', not only does there have to be a unitary way to 'win' – the cisgender norm – but further, she and her listener both have to make effort to change their speech towards this norm. This backed up in her behavior, as well as the behavior of the Cis CP; both of them shift towards fronter /s/ pronunciations with each other. In her conversation with Mantis, the Cis CP's mean /s/ CoG is 8009Hz, significantly different from her mean /s/ CoG with the other speakers (6642Hz; $p < 0.0001$). Here, the model of accomodation cannot provide a particularly clear analysis, as it is unclear who is accomodating to whom; analysis through the lens of Mantis' expressed understanding shows greater explanatory power.

## 3.2 Speaker 4: Jellyfish

Jellyfish, like Mantis, shifted towards fronter /s/ pronunciations when talking to the Cis CP, as shown in figure 3.3. Compared to Mantis, however, Jellyfish's upwards shift is smaller, with a difference in mean CoG of 300hz, and her overall mean CoG of at 6154hz is lower, closer to Squirrel and Alpaca. In my interview with her, Jellyfish emphasized the importance of lesbianism to her identity and her relationship with her voice, saying that "being a lesbian impacts the way that [she] move[s] through the world as a woman more than being trans does in a lot of situations," and that "[she] can sound like a boy as a lesbian, with the way that [she] present[s], and have people just be like 'oh that's like a weird dyke who smokes cigarettes'," which she considers to be an accurate description of herself. In this way, it makes sense to read her mean /s/ CoG as similar to that of the lesbian

[1] 'Cishet' is a shortened form of 'cisgender heterosexual'.

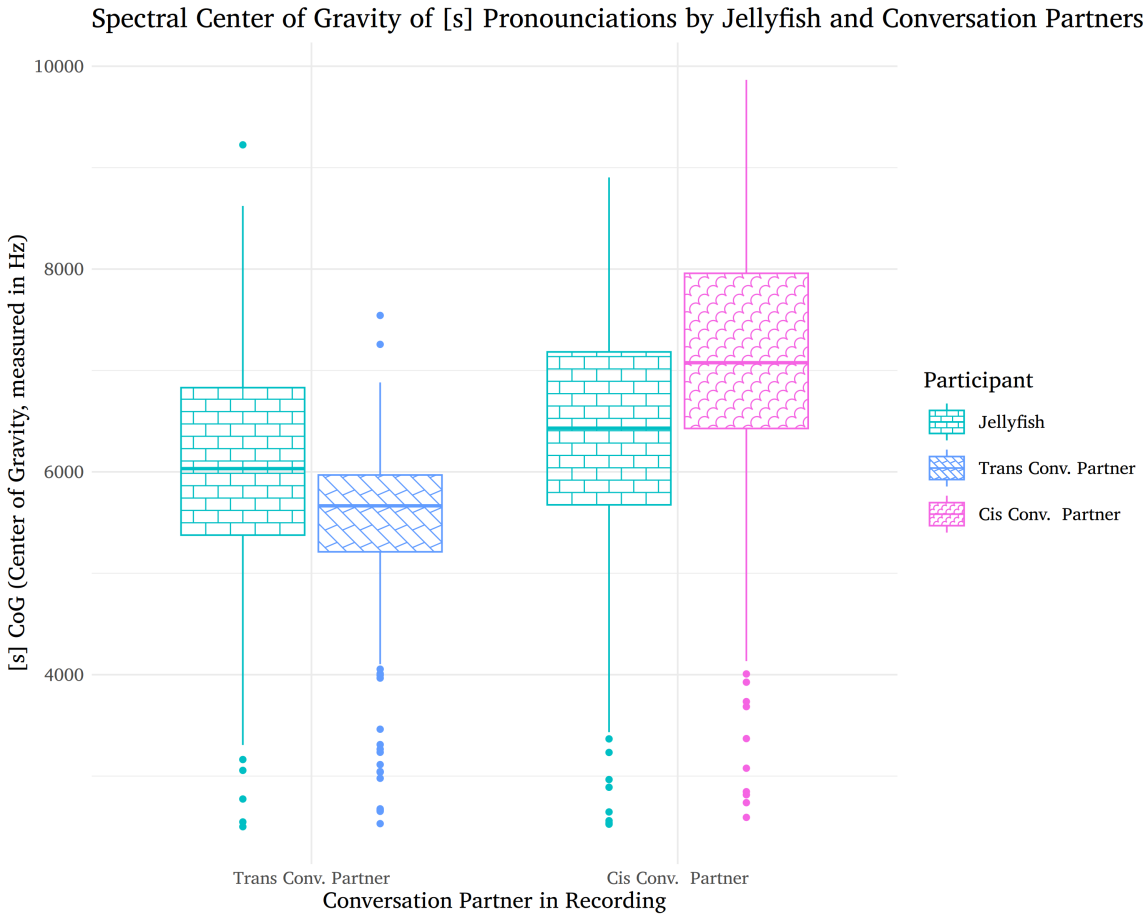



Figure 3.3: CoG of /s/ pronunciations in each recording containing Jellyfish, her pronunciations and the pronunciations of the conversation partners.

speakers Podesva and Van Hofwegen (2015) studied in Redding, California whose mean /s/ CoG was 6300Hz.

Similar to Mantis, Jellyfish lays out her transition as a process of increasing confidence in her gender and increasing ability to resist or cast off hegemonic feminine norms. For Jellyfish, this takes the shape of the development of a lesbian boy identity against earlier internalizations of ideological transphobia. She notes:

> "When I was in high school ... I would like look at all this like TERF[2] propaganda and more or less like fashion myself to be the exception to like the narrative around trans people."

She later notes that this caused her to become "kind of a psychopath, about like, presentation", paying obsessive attention at that time to "do like everything possible to fool people into thinking [she] was a cis woman". This was also the period where she notes:

[2]'TERF' is an acronym for 'trans-exclusionary radical feminist', a label for a transphobic ideology which attempts to justify itself through appropriation of feminism (see the discussion of gender-critical feminism and TERFism in Barrett and Hall 2024).

> "I realized that women whistle on their 's's way more, and I remember like consciously changing how I speak to like whistle on the 's' more."

This level of awareness of /s/ fronting is striking, and very interesting on its own. But even as an example of heightened awareness of hegemonic feminine norms and concern with having a 'passing' voice, this can be read as providing evidence for my prediction that speakers who report greater pressure to conform and pass through their voice will shift to fronter /s/ with the Cis CP. However, it is important to note that Jellyfish situates this quite clearly as an aspect of her past more than her present.

By contrast, Jellyfish's present is characterized by an attitude of resistance towards rigid conformance to a singularly feminine gender and expression. She expresses this confidently:

> " I don't think it benefits me in any sort of way to like restrict myself like that... people are way to serious about all this shit, ... like, if I'm dressed more masculine in a certain day I'll like listen to different music and I'll like talk differently and I'll move differently and I just think that's fun!"

When considered in light of my prediction that speakers who express attitudes of resistance towards hegemonic feminine norms will not shift to a fronter /s/ with the Cis CP, Jellyfish's outlook here perhaps raises the possibility a view of shifting to a fronter /s/ not as a conformist move towards the hegemonic norm, but rather as a shift in one's desired presentation, or a more neutral form of accomodation, motivated by a positive stance towards those around her. This view is also supported by her comment that she ""love/s/ talking like a boy when [she's] interacting with other boys"; here, accomodation may be read as a sort of harmonizing gender fluidity, taking up the gender of those around her.

Like Mantis, Jellyfish serves as a counter-example against my prediction that speakers who expressed concern with misrecognition would not shift /s/. Her discussion of this concern provides additional insights. She presents the issue of 'sounding like a gay guy' as particularly a concern in interaction with cis women, and as an unintentional habit, noting that:

> "It is like, annoying at times when I like, default back to talking like a gay guy around cis women ... more than sounding like a boy sounding like a gay guy is like, terrifying, because there's this relationship that, it's mostly cis women, have with like trans women where you will be accepted and tolerated as like a gay guy but anything beyond that, if you like, don't know your place more or less, you're getting into dangerous territory."

Jellyfish clarifies here that 'sounding like a gay guy' is not only a concern because of the inaccuracy and denial of her gender, but further that it represents a dangerous type of transphobia which hides itself behind an apparent friendliness towards cisgender queer people. In the phrasing 'default back to', she also calls up her

history. She mentioned in our interview that she identified as a gay boy for a period of time before realizing she was trans, and here she is perhaps indicating that 'talking like a gay guy' is a habit she developed during that time that she no longer likes but still deals with. I discuss the issue of listeners' personal history in relation to the question of gay male recognition as well as issue of intentionality and difficulty of changing how one speaks further in the next chapter.

## 3.3 Speaker 2: Alpaca

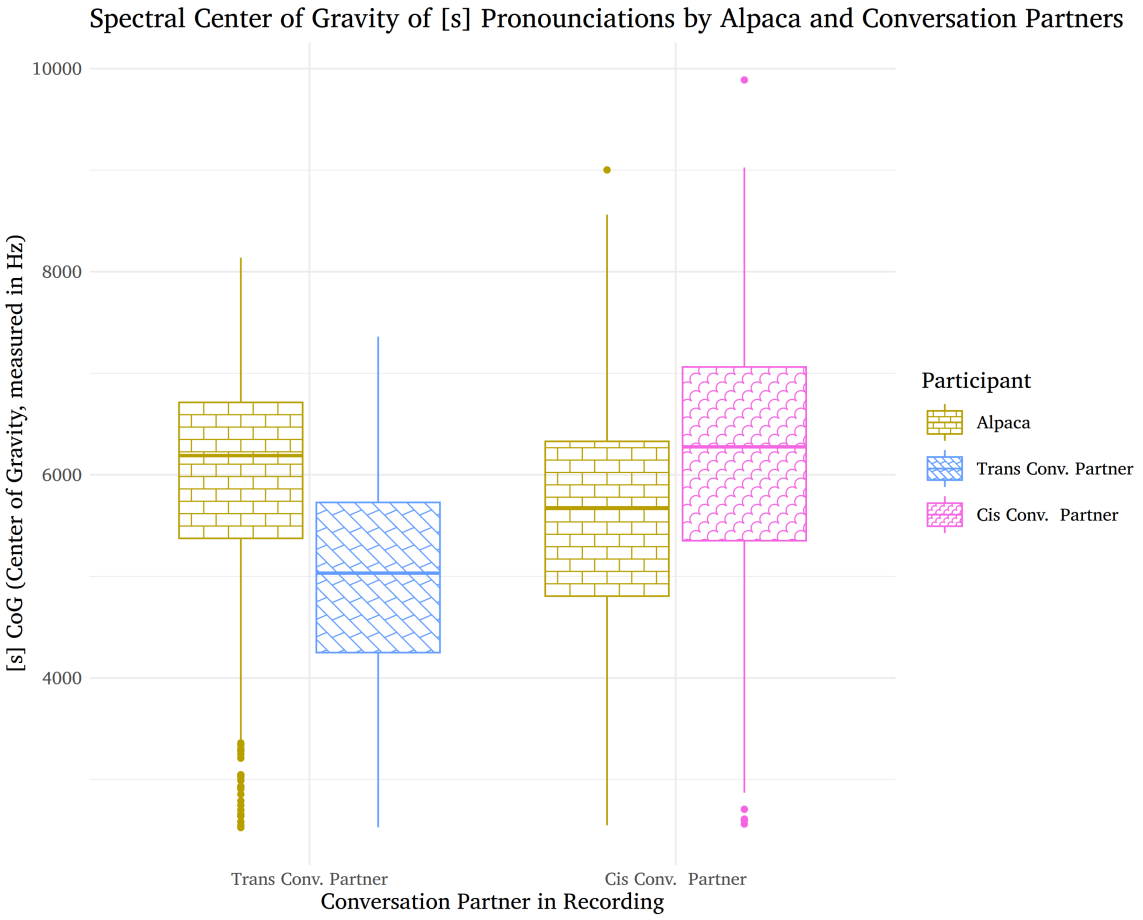



Figure 3.4: CoG of /s/ pronunciations in each recording containing Alpaca, her pronunciations and the pronunciations of the conversation partners.

Alpaca's /s/ pronunciations are shown in 3.4. Alpaca differed from Mantis and Jellyfish firstly in that she shifted her /s/ pronunciations to be backer with the Cis CP, rather than fronter; her CoG reduces by 400Hz. Also notable is that her overall average CoG of 5758Hz is the second lowest of my participants, only higher than the Trans CP's (5133Hz).

Also unlike Mantis and Jellyfish, Alpaca did not mention concern with sounding like a gay man in my interview with her. It is possible that her backer /s/

pronunciations are part of the reason that this concern does not come up for her, as her /s/ pronunciations are not front enough to trigger a misrecognizing gay-male indexicality. However, it is also possible that this concern is present, and simply went unmentioned in the interview. In this case, her shift to backer /s/ pronunciations could be to avoid this misrecognition. As either of these explanations are possible, my data on Alpaca is inconclusive with respect to the issue of misrecognition.

With respect to relationship with hegemonic femininity, the other factor my predictions associate with a lack of fronting around the Cis CP, Alpaca also differed from Mantis and Jellyfish. For Mantis and Jellyfish, the development of gender confidence involved a movement away from an initial feeling of pressure to strictly adhere to hegemonic femininity, and a gradual reclamation of behaviors which may hold hegemonic masculine associations as valid parts of a woman's identity. Alpaca describes the progression of her understanding of her gender differently:

> "I think it's been a process of just sort of like getting more comfortable with femininity, because not wanting to be masculine, is, in my thought process, is kind of different from, ... coming to terms with the fact that I do like femininity and that I am comfortable with it and that it is something I can enjoy and be comfortable in and have as my own part of my identity ..."

In this way, it seems that Alpaca's journey has been more of a gradual move towards comfort with elements of trans-affirming femininity, rather than a gradual move away from strict adherence to hegemonic femininity. Similar to the case with gay-male misrecognition, it is possible that Alpaca does distance herself from hegemonic femininity and that she simply did not mention so as explicitly as Mantis and Jellyfish. However, because my data does not include expressions of this attitude, I am unable to easily claim that she shifted away from fronter /s/ pronunciations in a move of resistance against hegemonic femininity.

With respect to the degree of concern with passing around cis people, Alpaca expressed similar feelings to the other speakers. Although she does not word this as strongly as the prior two speakers, not associating frustration or danger with cis women's perceptions, she notes

> "when I'm around other trans people I kind of feel less pressure ... to try to present perfectly femininely, like I feel less pressure to like pass ... in class or around cis people I might try to have a more feminine voice."

Counter to my prediction, this 'pressure to pass' does not result in a shift towards fronter /s/ pronunciations with the Cis CP for Alpaca.

From the perspective of my three attitudinal predictions, I am unable to provide a solid explanation for Alpaca's shift to backer /s/ pronunciations with the Cis CP. However, there are two possible confounding factors which may contribute to an explanation. First, Alpaca notes that she was not completely sure whether the Cis

CP was cis. This was a result of my choice not to explicitly inform speakers of the trans status of the conversation partners, in order to reduce priming effects. Although she did suspect that the Cis CP may have been cis, she also believed it was possible that the Cis CP was a highly passing trans woman. As a possible confounding factor, this may also play a role in explaining Alpaca's difference in behavior from the other speakers. Second, unlike the other speaker participants, Alpaca was friends with the Cis CP, another possible confounding factor.

## 3.4 Speaker 3: Squirrel

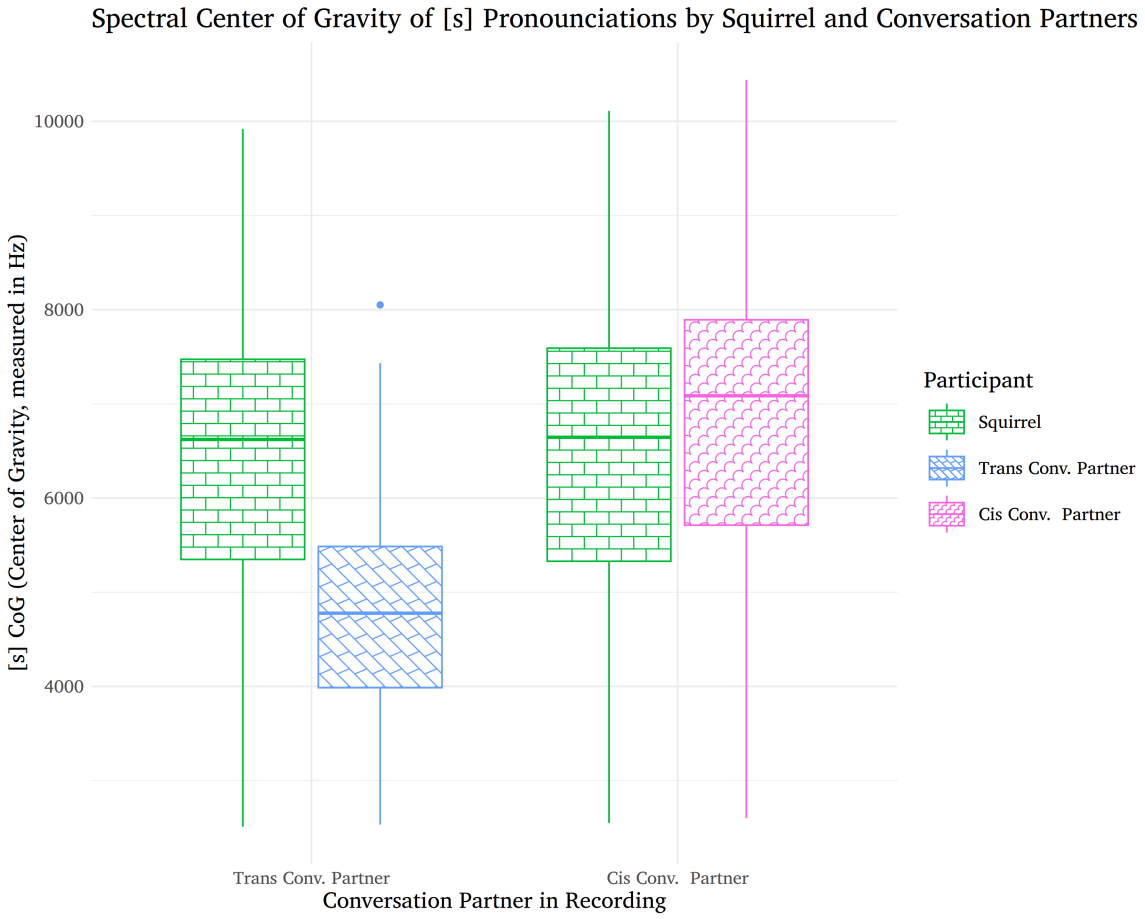



Figure 3.5: CoG of /s/ pronunciations in each recording containing Squirrel, her pronunciations and the pronunciations of the conversation partners.

Squirrel stands out as the only speaker who did not shift her /s/ pronunciations, maintaining a CoG of 6300Hz with both of the conversation partners (as shown in figure 3.5). She is also the only speaker in the early stages of her transition. She realized she was trans less than a year ago, and has not publically come out yet; only online friends and those close to her know her identity. In my interview with

her, she explained that she wished to physically transition through HRT before socially transitioning, and that she "didn't feel comfortable enough yet to" come out publically. She plans to come out next academic year, and intends to work on voice training in the summer before coming out. She did participate in this study in her transfeminine identity, and the CPs were aware that all of the people they would be talking with were transfeminine.

At this early stage of transition, it is perhaps surprising that her /s/ pronunciations are not backer, as she is the speaker participant with the second highest overall mean at 6357Hz. Although she has not yet begun efforts on voice training, it is possible she has already taken some less conscious steps towards a feminine voice, or that her /s/ pronunciation was already in this range pre-transition. In the interview, I asked her whether she felt that her voice was already feminine now, and she replied:

> "that's a complicated question... sometimes, sometimes not... when I was really young ... I used to be really insecure about my voice, but for the opposite reason, that I thought it was like too fem."

Although she cites a high overall voice pitch as the primary cause for these pre-transition insecurities, it is possible that /s/ pronunciations also played a role.

According to my predictions, speakers who hold greater attitudes of resistance towards hegemonic feminine norms and speakers who have concerns of misrecognition should be less likely to shift in /s/ pronunciations. From my interview with her, it seems that the former may be a factor in her choice not to shift. She notes that although she does intend to move towards a more feminine presentation,

> "I think a lot of times there's like this expectation of if you're transfem that you're like hyper-fem and that's not how I feel really personally, like I don't know if 'butch' is the right word for that but I definitely feel like the way that like I wanna present is not like super-hyper-fem."

Similar to Mantis and Jellyfish, Squirrel affirms here that she does not want to and does not have to strictly follow hegemonic feminine norms, and that she hopes to inhabit a femininity which may contain elements hegemonically associated with masculinity. She explains this through clothing, mentioning that she would prefer to wear all-black than a brightly colored sundress.

Squirrel does not mention concerns about being mistaken for a gay man with the sound of her voice. For Alpaca, I argued it was possible she was not concerned with gay-male misrecognition because of the backness of her /s/. This explanation is not possible for squirrel, as her /s/ pronunciations are fronter, within a similar range as reported for gay men (but also trans women) by Podesva and Van Hofwegen (2015). I discuss the implications of this mismatch in the following chapter. It is sufficient to note here that concern with misrecognition does not appear to be a factor in Squirrel's lack of shifting, counter to my predictions.

Squirrel's lack of shifting may also be interpreted as a counter-example to my prediction that speakers who express greater concern with sounding feminine and

passing would be more likely to shift, as it seemed that this concern was quite present for her. When I asked her about her relationship with her voice, she begun by explaining that in online voice chat interactions,

> "I like don't talk to a lot of people like that I would normally talk to if I was cis... as soon as I like basically came out to like my [online videogaming group]... I stopped talking entirely, on voice. ... I feel like it's easier for them to perceive me and like use correct pronouns like, if they don't hear me, because my, there's voice dysphoria, and like, my voice isn't like, they way I would want it to be if I could like pick it basically."

She explains further that

> "I feel like inherently when like you hear someone for the first time you kind of assign like gender in your head to like that person... if you hear a stereotypically more masc voice then you might like automatically start like 'he/him'ing someone in your head."

Here, she implicitly characterizes her voice in its current state as 'stereotypically more masc', in that she feels that not using her voice is the best way to avoid being perceived as a man in online interactions with cisgender people. For her, this falls under a general preference articulated as "the more ways I can like not perceived, the better." In this context, it is perhaps understandable that pressure to pass does not lead to a shift in /s/, as Squirrel perceives a passing or feminine voice as not yet in reach at her current stage of transition.

# Chapter 4

# Discussion

In this chapter, I discuss the above results in light of research question of this thesis raised above:

> Do transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener? What factors determine whether speakers shift?

The answer to the first portion of this question is relatively clear; Mantis and Jellyfish shift in this way. However, in order to be sure that this answer is not misleading, it is ncessariy to consider whether the speakers are speficially shifting in response to the trans status of the CP and the hegemonic indexical field of /s/, rather than in response to some other factor. Of primary concern here is the issue of accomodation; could these shifts be explained simply as the speakers pronouncing /s/ more similar to the CP they are talking with, rather than in reaction to whether the CP is cis or trans? The section that follows considers this question in detail. I find that although I cannot answer this question with certainty, it seems more likely, and more interpretively productive with respect to the second portion of the question (which factors determine whether speakers shift), that speakers are specifically reacting to the trans status of the CP and the hegemonic indexical field of /s/, rather than simply accomodating.

With respect to the second portion of the question – what factors determine whether speakers shift – I put forth an interpretation in which personal history of gay-masculine self perception emerges as the primary factor which differentiates the two speakers who shifted from those who didn't, reinterpreting the of gay-masculine misrecognition from my predictions. I then briefly discuss the role of race in my data, the failure of my other two predictions, and the implications of this data for the theories of misrecognition, the cisgender listening subject, and the hegemonic indexical field.

## 4.1 The Issue of Accomodation

Due to the small scale and scope of this study, my research design is unable to fully eliminate the factor of accomodation. In order to do so, it would either need to be the case that the two CPs have the same /s/ pronunciations – in which case accomodation could not result in any shift – or that more than one of each type of CP was included (e.g. two trans CPs and two cis CPs) – in which case shifts across listeners of differing trans status could be compared with shifts within listeners of the same trans status. Future research employing these designs could be highly illuminating. In the absence of these measures, it is certain that accomodation is taking place for Mantis and Jellyfish; their mean /s/ pronunciations fall in between the mean /s/ pronunciations of the CPs, and their shifts move them closer to the pronunciations of their CP. To point, though, is not whether these speakers are accomodating, but whether their motivations for their accomodation can be interpreted without reference to the trans status of the CP and the hegemonic indexical field for /s/.

How else, then, might theories of accomodation differentiate Mantis and Jellyfish as more motivated to accomodate than Alpaca and Squirrel? Research on sociolinguistic accommodation generally considers differences between individuals' propensity towards accomodation within two frameworks, neither of which, I argue, provide clear answers here. First, based upon patterns found within the cisgender heterosexual white population, it is often assumed that women accomodate more than men (e.g. Giles 2016). As all four of my speakers identify as women or girls, this generalization would thus expect them to be similarly inclined towards accomodation, and cannot differentiate between Mantis and Jellyfish. Second, accommodation is considered as a tactic of intersubjective adequation (e.g. Giles 2016), see Bucholtz and Hall 2005 on tactics of intersubjectivity), whereby speakers accomodate in order to create a sense of similarity and solidarity between each other. This conceptualization is more helpful in distinguishing why an individual accomodates more to one CP than another than in distinguishing why some individuals accomodate more than others to the same CPs. When the CPs are considered together, it is not clear that Mantis and Jellyfish were more interested in creating solidarity between themselves and the CPs than Alpaca and Squirrel were. This might expected if, say, there was some in-group to which the CPs, Mantis, and Jellyfish belonged, but Alpaca and Squirrel didn't. I am unaware of any way in which this could be the case; Mantis and Jellyfish did not seem to have any more in common with the CPs than Alpaca and Squirrel did.

At this point, the interpretation in which Mantis and Jellyfish are viewed as neutral accomodators, not reacting to the trans status of their CP or the hegemonic indexical field of /s/, ceases to provide helpful interpretations. Although I cannot reject the possibility outright – it is always possible there is factor outside of my data which could explain why Mantis and Jellyfish are simply more inclined to accomodate than Alpaca and Squirrel – I nonetheless find it unlikely that the trans status of their CPs and the hegemonic indexical field of /s/ are not relevant to these speakers' behavior, and find including these considerations interpretively

productive.

If speakers' /s/ shifting behavior is interpreted as responding to these factors, the answer to the first portion of the research question posed at the beginning of this chapter – whether transfeminine speakers shift towards fronter /s/ pronunciations in response to whether their listener is cisgender – is that Mantis and Jellyfish do. How does this approach, then, address the issue of why Mantis and Jellyfish shift to fronter /s/ with the Cis CP and Alpaca do not? I take speakers' commentary on the isssue of sounding gay-male as the primary explanatory factor to this question.

## 4.2 Motivations to Shift

In my original model, I considered Mantis and Jellyfish's expressions of concern with sounding like gay men as evidence that they would not front /s/ with a cisgender listener out of concern that /s/ may be interpreted as indexical of a gay man identity. This was not borne out, but it is nonetheless the case that Mantis and Jellyfish differ from from Alpaca and Squirrel on this issue (the former pair expressed these concerns, the latter pair did not). Here, I reconsider this issue in terms of personal history, exploring why speakers with a pre-transition history of a percieved gay-masculine voice might be more likely to front /s/ with cisgender listeners.

For Mantis and Jellyfish, I notice the issue of a percieved gay-masculine voice arises from these speakers' pasts. When asking speakers aobut their relationship with their voice and their gender, I framed the question in terms of histories, asking how these speakers relationships with their voices changed and developed throughout their lifetime. It is in this context that Mantis says that she has"always had like 'gay voice' or like, feminine voice"; the use of the word 'always' emphasizes that 'gay voice' was already a concern for her before transition. As I mention in Jellyfish's results section, she similarly situates 'talking like a gay guy' as something that she sometimes 'defaults back to'.

What do these speakers mean when they ascribe gay-masculinity to their voices? According to the body of research on the notion of gay speech, and according to Mantis' equivocation, 'gay voice' may simply be 'feminine voice' features in the context of a speaker or speech otherwise percieved as masculine, relying on the fractal-recursive structure of gendered and sexualitized sociolinguistic meanings. It is notable that these speakers apply these perceptions to themselves; they do not frame their discussions in terms of others' saying that they sound like gay men, but rather hearing themselves as gay men. Before transition, these self-perceptions were likely the result of noticing femininity within themselves despite an maintaining an assigned masculine identification, resulting in self-perceptions of gay-masculinity. After transition, dysphoria or internalized transphobia may result in continuing self-perceptions of masculinity, and the contrast between these perceptions and speakers' otherwise feminine presentations similarly result in gay-masculine self-perceptions. These self-misgendering moments are an example that

the cisgender listening subject is an ideological perspective promoted by structures of domination of cisgender people over transgender people, rather than a specific group of cisgender individuals. As a result, these speakers may apply the judgements of the cisgender listening subject to their own voices, seeing themselves through the marginalizing lens of hegemonic norms. In other words, that they sometimes feel upset that their voices sound gay, rather than feminine, is likely result of the internalization of hegemonic and marginalizing gender norms.

Although most language variables indexical of gay masculinity can also be indexical of hegemonically normative femininity, it is nonetheless the case that some variables are more well-known as indexes of gay speech. This is the case for /s/; there is a culturally recognized metalinguistic notion of the so-called 'gay lisp' **calder_from_2020**, which considers gay men's use of fronter /s/ pronunciations to sound like a 'lisp'. It is possible that when Mantis and Jellyfish hear their voices as gay-masculine, /s/ is of primary concern (consciously or unconsciously) due to its status as the 'most iconic' (**calder_from_2020**) index of gay sexuality. It is possible, then, that Mantis and Jellyfish may be more aware (consciously or unconsciously) than Alpaca and Squirrel that /s/ can index gay masculine speech (the $n+1$st indexical order), because they may have applied this indexicality to themselves in moments of dysphoric self-misrecognition. It is possible, then, that these speakears may also come to hold greater awareness of the $n$th indexical order of hegemonic indexicalities of fronted /s/ to femininity as well, and thus make greater use of these indexicalities in defending themselves against the possible judgements of the cisgender listening subject they predict the Cis CP may inhabit. In other words, being concerned with sounding gay raises speakers' awareness of the importance of /s/, which they then come to use to hegemonically index femininity to protect themselves against the judgements of the cisgender listening subject.

I consider this interpretation to provide the most unified explanation my data can provide for why Mantis and Jellyfish front /s/ with the Cis CP, and Squirrel and Alpaca do not. It calls for a revision of my prediction with regards to concern with gay-masculine misrecognition, in that it acknowledges that avoiding fronter /s/ pronunciations is unproductive for speakers. If speakers assume that their /s/ pronunciations will be interpreted through a male perceptual frame (within the fractal-recursive binary opposition at the $n+1$st indexical order), then the only options with /s/ are straight man and gay man, both of which are undesirable perceptions. Instead, the best option available to speakers who are aware (consciously or unconsciously) of the structure of hegemonic indexical meanings is to front /s/ in the (conscious or unconscious) hopes that their /s/ pronunciations will be interpreted through a female perceptual frame (within the fractal-recursive binary opposition at the $n$th indexical order) and contribute to overall feminine perceptions.

Because this interpretation is primarily concerned with the binary distinction between shifting to fronter /s/ pronunciations with the Cis CP (moving towards hegemonic femininity) and not doing so, it is unable to differentiate between Alpaca, who shifted to backer /s/ pronunciations with the Cis CP (away from hegemonic femininity), and Squirrel, who did not shift. With the data I have, I am

not able to provide a confident explanation for this difference. However, I suspect that the confounding factors of Alpaca's pre-existing familiarity with the Trans CP and uncertainty as to the trans status of the Cis CP likely play a role. Aside from the absence of a personal history of self-percieved 'gay voice' resulting in a less heightened awareness of the role of /s/ within hegemonic gender norms, Squirrel's transition stage seems additionally relevant to the absence of a shift. Even if she is aware (consciously or unconsciously) that her /s/ pronunciations may effect how her gender is percieved by a cisgender listener, it is possible she only has access to a single kind of /s/ pronunciations, as she has not had time to practice and learn to pronounce /s/ with a fronter tongue position. Zimman (2021b) demonstrates that a multi-dimensional model of gender is needed to explain the /s/ pronunciation patterns of his gender-diverse set of speakers, including gender assigned at birth, current gender identity, and gender role – the gender the speaker currently performs on a daily basis (which for Squirrel is, until she comes out, man). My data agrees with Zimman's model here, affirming the importance of considering gender role.

An additional interpretive note regards the factor of race in my data. Because the hegemonic indexical field for /s/ is based upon specifically white gender norms, It is possible that for Mantis, her shift worked to protect her from the judgements of the white listening subject as well; a number of studies of race and style-shifting have found that Black speakers shift towards hegemonically white normative speech with white listeners (eg. Rickford and Mcnair-Knox (1994)), and it is possible that Mantis, as an Asian-american, engaging in a similar practice here. That Mantis showed the widest shift can also be interpreted through the lens of race, in that with the Trans CP, who is Chinese, Mantis could speak without regard to both cisgender and white voice norms. In my literature review, I was unfortunately unable to find research which discusses Asian-Americans' /s/ pronunciations; research in this area could provide further insights here.

Above, I consider the primary factor of personal history as well as secondary factors of transition stage/gender role and race to explain whether speakers choose to shift. Under this interpretation, my data provides evidence for the relevance of the theory of cisgender white listening subject. In addition to its relevance to transfem thinking about the voice and being heard (Heffernan 2023) and its inhabitation by real cisgender listeners (Jessee and Calder 2025), it is clear that the influence of the cisgender white listening subject does, for Mantis and Jelllyfish, manifest in /s/ shifting. This study thus supports the usefulness of listening subjects as a theorization of the mediation between perception of speech and production of speech, especially for trans speakers.

## 4.3 Additional Observations: Resistance and Degrees of Control

In my initial model, in addition to the issue of gay-male misrecognition, which has been reinterpreted above, I made predictions regarding speakers' relationship with hegemonic femininity, as well as concern with cisgender perceptions. As discussed in the results section, all four speakers expressed these feelings in their own ways. They all noted how they might change how they speak around cisgender people and mentioned ways in which cisgender perceptions pressure them to conform or make them feel dysphoric. All four speakers expressed nuanced relationships with femininity and described how the process of transition has brought them to their own form of femininity, which may not adhere to hegemonic norms. In the interpretation I put forth here, I do not find these factors to determine whether speakers shift with /s/. This emphasizes the powerful position of the cisgender listening subject; under my interpretation, speakers prioritize protecting themselves against cisgender perceptions, and and shift /s/ if they have sufficient (conscious or unconscious) awareness of the importance of /s/ within hegemonic gendered speech norms.

Although it is possible that Squirrel and Alpaca's shifts could be read through the lens of resistance, where the risk of negative cisgender perceptions is accepted in exchange for the freedom of casting off hegemonic norms, I am hesitant to make this conclusion. As mentioned above, all speakers express discontent with hegemonic norms and cisgender perceptions, and I have no reason to believe that some speakers would be more resistive than others. Additionally, I do not wish to imply that Mantis and Jellyfish's shifts could not be considered resistance; they are resisting the cisgender listening subject by taking up /s/ as a hegemonic indexical resource, refusing to allow themselves to be misrecognized. Furthermore, it is difficult to determine whether speakers are primarily moving towards or moving away from hegemonic speech norms when only investigating a single form. Future research which considers /s/ as one element of a constellation of indexically-relevant sites of speech variation, including, for example, overall pitch (f0), intonation, voice quality, or discourse markers could resolve these questions. For the moment, I am unable to conclude whether my speakers should be said to be resisting, and hesitate to make the judgements which may be required to make this sort of determination.

Nonetheless, future research carefully considers how trans and otherwise marginalized speakers implement social-indexical resistance could be illuminating; Barrett and Hall (2024) mention the ways in which marginalized groups can take advantage of misrecognition resistively or recontextualize hegemonic indexicals into subverting contexts (termed indexical disjuncture). Sociolinguistic research which investigates more closely the relationship between speakers' resistive intentions and their linguistic behavior, as I attempt to somewhat inconclusively here, could illuminate how speakers balance between self-protection and defiance of norms.

An additional factor which may be relevant to my data concerns intentionality

and accessibility of shifting. In my interviews with my speakers, they characterized their voices as not always and completely under their own control, and that this control requires some practice to develop. As mentioned above, Jellyfish characterises 'talking like a gay guy' as a type of 'default'ing. She uses the metaphor of rolling dice to describe her relationship with various types of voices she feels she has access to,

> "It's kind of rolling the dice what voice comes out on a certain day, and sometimes the faggcent comes back"

and considers 'talking like a boy' to be her neutral state:

> "at a certain point, I like stop performing and sort of default back to like my neutral state which is still just like talking like a boy, which is fine, like I don't really feel much pressure to change that."

Nonetheless, she also expresses that she is proud of her ability to styleshift, and that this grants her freedom:

> "I'm pretty proud of my voice, and like, um, being able to more or less manipulate it how I want, um, and, I feel like it gives you a lot of freedom in like how you move through the world if you can like talk in different ways to different people."

The overall picture that is drawn here is that Jellyfish considers styleshifting, and controlling her voice in general, to be a learned skill, rather than an inherent ability, and attributes times when her voice is not the way she would like it to be to the lack of complete control.

My other speakers view their relationships with their voices similarly. Squirrel mentions that her voice is "not the way [she] would want it to be if [she] could like, pick it" when describing her voice dysphoria. Alpaca mentions that she learned "the exercise to raise your larynx ... and then [she] found [her]self sort of subconsciously doing it" over time; styleshifting here must be learned, in Alpaca's case, consciously – (although it would still require learning unconsciously). For Mantis, her experiences with voice dysphoria, but also with passing with her voice, similarly draw a picture of partial, rather than complete, control over her voice.

I find this tendency notable because it is somewhat out of alignment with the general assumption of modern frameworks of social-indexical linguistics, that speakers' speech patterns should be understood as intentional and aligned with their goals. In the interpretation of my results put forth above, I consider the insights my speakers provide about their partial control over their voices helpful. In order for speakers to be able to take up the move of self-defense against the cisgender listening subject with /s/, they have to understand the importance of /s/ as an indexical resource and have practice with changing their /s/ pronunciations. I therefore encourage future sociolinguistics research to take up the issue of intentionality with nuance, as I attempt to do here, looking at speakers' actions through the lens of intentionality first, aiming to find ways in which speakers take up indexicality in line with their goals, but secondarily considering this issue of access when intentionality and behavior seem out of alignment.

# Conclusion

In conducting an comparison of transfeminine speakers' /s/ pronunciation with a cis listener and a trans listener, this thesis aims to shed light on the ways that marginalized individuals navigate norms which exclude and restrict them. The field of language and gender is well-suited to this investigation, as it provides in-depth understandings of linguistic gender norms through close analysis of perception and perception of social meanings through language use. However, research on trans speakers has primarily taken two approaches to hegemonic gender norms; it has either compared trans speakers against cis speakers and characterised their differences from these norms, or analysed the unique patterns of language use constructed within trans-safe communities. By investigating directly the effect that the presence of cisgender listeners has on transfeminine speech, I aim to bridge this gap, answering two primary theoretical questions present in the existing research on marginalized speakers.

First, in response to a growing body of research on priviliged listening subjects – ideological positions which apply hegemonic, marginalizing understandings in the interpretation of sociolinguistic signs – I ask how speakers react to the presence of a listener who may inhabit a priviliged listening subject position in actual linguistic practice. Second, I ask how speakers model their listeners' indexical inventories, aiming to understand the mediation between linguistic perception and linguistic production. I find that two speakers move towards fronter /s/ pronunciations with the Cis CP and two do not, and interpret the implications of my findings for these questions.

With respect to the first question, although I am unable to prove with certainty that my speakers' shifts towards fronter /s/ with the Cis CP are not simply a matter of accomodation, I find that considering their shifts as reacting to a percieved presence of the cisgender listening subject produces a stronger interpretation. In combination with my finding that my speakers construct the cisgender listening subject within their metalinguistic commentaries (similarly to Heffernan 2023), I argue that transfems' concept of the cisgender listening subject influences their actual linguistic practice. I consider the speakers who shift towards fronter /s/ pronunciations to be protecting themselves from the possibility of misgendering judgements by moving into greater alignment with /s/ pronunciations considered to be feminine.

In considering why only two of my four speakers shift this way, although I am similarly unable to explain this difference with certainty, I notice that the speakers

who shifted discuss a history of self-percieptions of their voice as sounding gay masculine. The indexical link present in the hegemonic indexical field from fronted /s/ pronunciations to gay men's speech is highly salient (**calder_from_2020**). I find it likely that speakers who are more concerned with sounding gay-masculine will thus hold greater (conscious or unconscious) awareness of this indexical link. It is thus possible that these speakers will place greater weight on /s/ as an index of femininity than than others, and are more likely to take up /s/ to protect themselves from the judgements of the cisgender listener.

With respect to the second question posed above, I conclude that marginalized speakers model the indexical field of priviliged listeners through complex and complete models of the hegemonic indexical field. In my interpretation, speakers' model the hegemonic indexical field of /s/ in line with the fractal-recursive indexical order model put forth by Calder (2019a); /s/ enters my speakers' awareness through its salient meaning of gay-masculinity at the $n+1$st indexical order, but this indexicality brings with it the binary-genderred indexical links to masculininity and femininity on the $n$th indexical order. As an example of the close interdependence of indexical orders (see Silverstein 2003), it can be seen that marginalized speakers come to a complete (conscious or unconscioused) understanding of the hegemonic indexical field, and respond in consideration of this full structure.

In light of the theory of indexical misrecognition put forth by Barrett and Hall (2024), I see that marginalized speakers can employ generalized understandings of hegemonic meanings to model the indexical inventories of specific priviliged listeners. Here, speakers act under the assumption that these listeners will apply the hegemonic field to all speakers, an act of indexical presumption, and protect themselves from indexical misrecognition accordingly.

Taken as a whole, my study hopes to simultaneously highlight the difficulty of operating within a system of exclusive and marginalizing norms and the freedom speakers can build to navigate those norms as they choose. As my participant Jellyfish says,

> "Fundamentally, queerness should change how you move through the world. I think that's a good thing, I think like having control of how you're perceived and how you interact with the world and thinking about that more is something that everyone should do, and my transness and my lesbianism are like vectors to which I do that."

## 5.4 Limitations and Directions for Future Research

A primary limitation of this study relates to its lack of raciolinguistic insights. As a white researcher working with mostly white participants and drawing on a mostly white-centric research literature, I perpetuate the whiteness of trans studies, queer studies, and sociolinguistics. I have attempted to point out the racial dynamics in my data as I can see them, and to contextualize the whiteness of the norms I investigate, to mitigate this issue and situate my research within its speficic racial context.

Given the whiteness of the hegemonic gender norms around /s/ (A. J. Steele 2022; J. Calder and King 2022), it would be particularly illuminating to investigate BIPoC transfeminine speakers' /s/ pronunciations. A study with a similar methodology as mine within this population may find that that race of the perceiver is equally relevant as cisgender status, and that the influence of the white cisgender listening subject would exert multiplicative oppressive effects, an example of sociolinguistic intersectionality (see miles-hercules 2022). I encourage future research to specifically focus outside white speaker and perceiver populations, and hope to be able to work with queer and trans BIPoC sociolinguists and speakers to focus closer on race in the future.

Other limitations of this study pertain to scale and scope. As mentioned in my discussion, this study could greatly benefit from more angles of analysis in several ways. To name two, expanding to include multiple linguistic variables alongside /s/ would provide much-needed context about how speakers engage with hegemonic norms more broadly, and zooming in to the discourse level to inspect how /s/ was shifted within each individual conversation based on topic, stance, and mood would allow a more nuanced understanding of the moment-to-moment meaning-in-context of /s/ as it develops and shifts, which is lost in the aggregated understanding I present. Further research would also benefit significantly from a larger number of participants, at which a greater diversity of participants and conversation partners could reveal broader patterns and reduce the possible confounds introduced by any idiosyncrasies of the individual personalities of the two conversation partners I have. This would also allow for a quantitative analysis of predictive factors of /s/ shifting, operationalizing with more precision the attitudes, concerns, and identities I consider qualitatively here.

# Appendix A

# Survey

# Appendix B

# Interview Questions

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